

CHAPTER EIGHT

PRICING UNDER IMPERFECT COMPETITION

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8.1. Introduction and learning objectives

Under perfect competition the actions of an individual agent are irrelevant to all other economic agents in the same market. No single agent has direct influence on the overall result (e.g. prices) on the market; the term ‘atomistic market’ is sometimes used to describe this situation. Perfect competition is intended less as a description of actual markets, and more as a thought experiment that describes an extreme situation. In the vast majority of agri-food markets, competition is imperfect and the actions of individual firms typically play an essential role in determining the market outcome.

Numerous examples from all stages of the agri-food value chain illustrate this. Global production of mineral fertilisers, for instance, is dominated by about ten internationally active companies. High development costs for new seeds are a major reason for increasing market concentration in the field of plant breeding; here, too, only a few firms are active worldwide. The process of concentration is particularly pronounced in food retailing. Retail chains may be able to use their bargaining power in negotiations with food processors. In addition, the high concentration in food retail in many countries raises the question whether consumers pay too much for food.

In this chapter we discuss:

- how prices are determined in different types of markets and the consequences of imperfect competition (market power) for the prices of goods,
- how different business strategies affect market equilibrium (in the short and long term),

- what factors influence the intensity of competition in a market, and under what conditions collusion between companies is likely to take place,
- why the diversity of products in the agri-food industry has increased so much,
- what factors influence product diversity in a market and how food diversity is related to imperfect competition,
- how the market power of companies can be measured, and what practical problems arise when measuring market power,
- if and how businesses can be prevented from exercising market power, and
- how competition policy works in Europe and what problems it entails.

As mentioned above, under perfect competition the prices of goods are exogenous from the point of view of the individual suppliers; firms will therefore aim to produce goods at minimum cost. In markets with imperfect competition, the decision-making of firms is more complex, because a single firm can influence the market price through its behaviour. In deciding on the optimal price and whether or not to change its price, two issues are particularly relevant for a firm: (a) how will consumers react to a price (change), and (b) how will competitors react. Before we explore strategic interaction between firms in more detail, we will first briefly discuss consumer behaviour.

8.2. Consumer behaviour

“Essentially, all models are wrong, but some are useful”.
Box und Draper (1987, p. 424).

Knowledge of consumer behaviour is essential for corporate decision-making, and also for the economic analysis of businesses’ pricing decisions under imperfect competition. Due to its importance, a vast economic literature has accumulated on consumer behaviour. We cannot provide a comprehensive treatment here; instead, to provide a basis for the subsequent

analysis of corporate strategies, we will only present a simple model of the demand for differentiated products.¹

We begin by assuming that a representative consumer's preferences for the n goods of interest are characterised by a quadratic utility function (Häckner 2000):

$$U(q, q_0) = \sum_{i=1}^n s_i q_i - \frac{1}{2} \left(\sum_{i=1}^n q_i^2 - 2d \sum_{j=1}^n q_i q_j \right) + q_0 \quad \text{for } i \neq j \quad (1)$$

where q_i is the quantity of the i -th good consumed, and n is the number of different goods. This utility function also includes the utility of other products consumed, labelled q_0 , a composite commodity that is assumed to be independent of the n goods (i.e. it is neither a complement nor a substitute to these n goods). q_0 is assumed to be the numeraire (i.e. its price is set equal to one: $p_0 = 1$). The parameter $d \in [-1, 1]$ measures the degree of horizontal differentiation between products i and j . For our purposes it does not matter whether i and j are objectively different (i.e. have different physical characteristics) or are only subjectively judged to be different by the representative consumer.² If $d = 0$, the products are independent, and each firm enjoys monopoly power for its product. $d = -1$ describes the case of perfect substitutes (the products are not differentiated at all), while $d > 0$ describes a complementary relationship between the products. Finally, the parameter s_i as a measure of product quality, also known as vertical differentiation. *Ceteris paribus*, an increase in s_i leads to an increase in the marginal utility of consuming one unit of product i .

Linear demand functions for each of the n goods are obtained by maximising the utility in equation (1) subject to the budget constraint $y = \sum_{i=1}^n p_i q_i + q_0$ (where y is income – see Appendix 8.1 for details):

¹ A comprehensive account of the theory and empirical analysis of demand behaviour can be found in Deaton (1986). Angus Deaton received the 2015 Alfred Nobel Memorial Prize in Economics "for his analysis of consumption, poverty and welfare" (Nobelprize.org 2020).

² Note that the model assumes that the degree of substitutability d is identical between all products. In other words, if the products are different, they are all equally different from one another. This is a major limitation of this simple model.

$$p_i = s_i - q_i + d \sum_{j=1}^n q_j \quad (2)$$

$$\text{or} \quad q_i = \sigma_i - \beta p_i + \delta \sum_{j=1}^n p_j \quad \text{for } j \neq i \quad (3)$$

We obtain the demand function in equation (3) by rearranging the inverse demand function in equation (2) and defining the following parameters:

$$\sigma_i = \frac{[1 - d(n-2)]s_i + d \sum_{j=1}^n s_j}{(1+d)[1 - d(n-1)]}, \quad \beta = \frac{1 - d(n-2)}{(1+d)[1 - d(n-1)]}$$

$$\text{and} \quad \delta = \frac{-d}{(1+d)[1 - d(n-1)]} \quad (4)$$

If the goods are independent ($d = 0$), the quantity of good i demanded only depends on its own price p_i and not on the prices of other goods p_j (because $\sigma_i = s_i$, $\beta = 1$ and $\delta = 0$). If the goods are perfect substitutes ($d \rightarrow -1$), then $\beta \rightarrow \infty$ and $\delta \rightarrow \infty$ in equation (3), which indicates that consumers react with infinite price-sensitivity to changes in the prices of both goods. In the more common case of imperfect substitutes ($-1 < d < 0$), which we will consider in greater detail below, we obtain $\beta > \delta > 0$ (for $n \geq 2$).

The influence of quality differences between goods is represented by the parameter σ_i . If the n substitutes are of the same quality (i.e. the goods are differentiated horizontally but not vertically), then $s_i = s_j$ and $\sigma = \frac{s}{1-d(n-1)} > 0$. If the goods differ in quality, σ_i increases with the quality of good i (s_i), and decreases with the quality of competing goods (s_j). If company i succeeds in increasing the perceived (subjective) quality of its own good (s_i) through advertising, σ_i and the demand for its own good q_i in equation (3) increase, while σ_j and demand for the competing product q_j decrease. This case is referred to as **predatory advertising**. By contrast,

cooperative advertising describes a positive relationship between the advertising expenditure for good i and the demand for other goods j .³

The demand model described above is limited, e.g., by the assumption of a quadratic utility function, and by the assumption of identical substitutability d between all goods. Nevertheless, the model is comparatively simple and provides a sufficient basis for the following analysis of strategic interaction between suppliers of differentiated goods.

8.3. Short-term strategic interaction in a duopoly

Markets with imperfect competition are characterised by the fact that economic agents interact strategically: the behaviour of one actor influences the behaviour of another, and *vice versa*. Even very large suppliers with large market shares pay close attention to what their competitors are doing and adjust their own behaviour accordingly. In this regard, markets with imperfect competition differ from monopolies and competitive markets.⁴

In general, we distinguish between two types of competition (i.e. types of strategic interdependence): Competition with **strategic substitutes** and competition with **strategic complements**.⁵ If x_i is an action by an economic agent i (e.g. to produce a certain quantity, to set a particular price, or to spend a certain amount of money on advertising), either $\frac{\partial x_j}{\partial x_i} < 0$ (strategic substitutes, the actions offset) or $\frac{\partial x_j}{\partial x_i} > 0$ (strategic complements, the actions reinforce one another) will apply. As we demonstrate in the next section, when firms compete with one another by adjusting the quantities of goods that they sell, their actions can be characterised as strategic substitutes. When firms compete with one another on the basis of price their

³ Whether advertising actually is predatory or cooperative is case-specific. Gasmi et al. (1992) study the market for Pepsi Cola and Coca-Cola in the US and find that advertising expenditures in this market have predatory effects. According to Rojas (2008) advertising expenditures in the American beer market have predominantly cooperative effects.

⁴ We assume that the reader is familiar with the economic analysis and consequences of monopoly behaviour. Appendix 8.2 contains a brief refresher on monopoly behaviour in a model with two goods.

⁵ The terms ‘strategic substitutes’ and ‘strategic complements’ refer to relationships between the actions of companies and should not be confused with the terms ‘substitute’ and ‘complement’ as they refer to relationships between goods from the perspective of customers.

actions are typically strategic complements. Which type of competition prevails has important implications for the choice of optimal long-run strategic decisions.

8.3.1. Competition with strategic complements (price competition)

In the short term, pricing is one of the most important business decisions. To determine the profit-maximizing price for a particular product, firms must consider three factors: the consumers' willingness to pay, production costs, and competitors' reactions. To illustrate this with a simple model, we begin by considering a duopoly ($n = 2$). Each firm produces one of two differentiated products, which they produce at constant marginal costs ($c'_i \equiv \frac{\partial c_i(q_i)}{\partial q_i}$) and fixed costs (f_i). What price should a firm ($i = 1, 2$) choose if it aims to maximise its profit and must decide simultaneously (i.e. it cannot observe its competitor's price before deciding)? The solution to this question is closely linked to the name of the French mathematician Joseph **Bertrand**, who was the first to analyse price competition in a duopoly.

Given the demand equation (3), the profit maximisation problem of the first firm ($i = 1$) is:

$$\max_{p_1} \pi_1 = (p_1 - c'_1)q_1 - f_1 = (p_1 - c'_1)(\sigma_1 - \beta p_1 + \delta p_2) - f_1.$$

The first-order condition for a profit maximum is:

$$\frac{\partial \pi_1}{\partial p_1} = \sigma_1 - 2\beta p_1 + \delta p_2 + \beta c'_1 = 0.$$

Hence, the first firm's profit-maximising price must satisfy the following first-order condition:

$$p_1 = \frac{\sigma_1 + \delta p_2 + \beta c'_1}{2\beta} \equiv R_1 \quad (5)$$

Equation (5) shows that the first firm's optimal (profit-maximising) price depends on the parameters of the demand function (σ_1 , δ and β), on the firm's own marginal costs c'_1 , and on the competitor's behaviour (expressed by the price p_2). Since p_2 is not observable when the two firms decide simultaneously, firm 1 must react to the expected price p_2 . A firm's optimal pricing rule as a function of its competitor's behaviour is referred to as its

reaction function or best response function (R_1). The reaction function in equation (5) shows that the first firm's profit-maximising reaction to a price increase by its competitor is to increase its own price: $\frac{\partial p_1}{\partial p_2} = \frac{\delta}{2\beta}$, where $0 < \frac{\delta}{2\beta} < 1$ (since $\beta, \delta > 0$ and $0 < \delta < \beta$ for $n \geq 2$ follow from equation 4). If the price of product 2 increases, then firm 1 will expect an increase in demand for its own product 1, and will therefore increase the price of this product. Since action (p_2) and reaction (p_1) reinforce one another, prices are strategic complements if the goods being considered are substitutes (i.e. $d < 0$ and therefore $\delta > 0$).

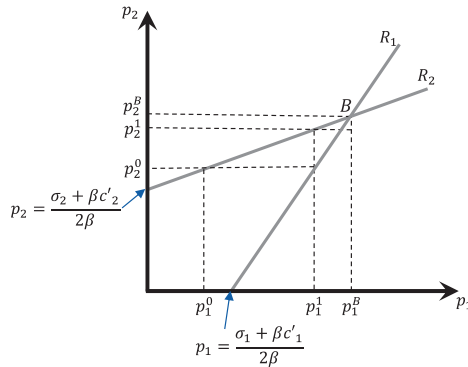
We can describe the second supplier's situation in the same way. Firm 2's optimal pricing decision will also depend on demand parameters, production costs, and the competitor's expected behaviour. Maximising profits for firm 2 leads to:

$$p_2 = \frac{\sigma_2 + \delta p_1 + \beta c'_2}{2\beta} \equiv R_2 \quad (6)$$

Given the behaviour that is described by the reaction functions in equations (5) and (6), does an equilibrium exist in this market and, if so, what prices are chosen for the two goods? To answer this question and analyse the resulting behaviour, a graphical representation is useful.

Figure 8.1 shows the reaction functions for the two firms. As mentioned above, firm i reacts to a price increase by its competitor j by also increasing the price of their own product, with $0 < \frac{\partial p_i}{\partial p_j} < 1$. For example, suppose the first firm considers setting a price of p_1^0 for its product. Whether this price is the best possible choice for firm 1 will depend, among other things, on how firm 2 responds to that price. Figure 8.1 shows that the best response by firm 2 is to choose a price of p_2^0 on its response function R_2 . If firm 1 correctly anticipates this reaction, it will conclude that p_1^0 is not the best possible decision. If the second firm were indeed to respond by choosing p_2^0 , then according to the response function R_1 the first firm's best choice would be p_1^1 , not p_1^0 .

Figure 8.1: Reaction functions in a duopoly with price competition



Source: Own depiction

The first firm will evaluate all possible price choices based on similar considerations. For each price that it could choose, it will try to anticipate its competitor's reaction and then, in light of this reaction, evaluate whether its original choice was optimal. These considerations will lead the first firm to the conclusion that there is only one price that maximises its own profits given a profit-maximising response by the second firm. This price is p_1^B . Firm 2 will respond to this price with p_2^B . If the first firm correctly anticipates this, it will conclude that selecting price p_1^B was the best possible decision and that there is no way to further increase profits, given the second firm's behaviour (i.e. p_2^B).

The second firm will conduct analogous thought experiments. In this way, it will also reach the conclusion that the price p_2^B maximises its own profits given its competitor's optimal reaction. The price combination (p_1^B, p_2^B) is the only combination for which neither competitor wishes to revise its pricing decision, given the other competitor's anticipated behaviour; at these prices there is no ex post regret. The first firm's decision to choose p_1^B is optimal given that the competitor chooses p_2^B , and vice versa. This form of equilibrium is called a **Nash-Bertrand equilibrium**.⁶

⁶ John Forbes Nash (1928 - 2015) was an American mathematician who, together with Reinhard Selten and John Harsanyi, received the Alfred Nobel Memorial Prize in Economics in 1994. His story became known to a wider audience in 2001 through the film "A Beautiful Mind".

In our example, the equilibrium can be calculated by solving equations (5) and (6) (R_1 and R_2):

$$p_i^B = \frac{2\beta(\sigma_i + \beta c'_i) + \delta(\sigma_j + \beta c'_j)}{4\beta^2 - \delta^2} \quad \text{for } i, j = 1, 2 \text{ and } i \neq j \quad (7)$$

In the simple case of symmetric demand ($\sigma_1 = \sigma_2 = \sigma$) and identical marginal costs ($c'_1 = c'_2 = c'$) the two firms will choose the (same) price $p^B = \frac{\sigma + \beta c'}{2\beta - \delta}$. The equilibrium price of a product will increase as consumers' willingness to pay or product quality (σ) increase, as consumer sensitivity (β) decreases, and as the marginal cost of production (c') increases. Furthermore, the degree of product differentiation plays an important role: the more differentiated the products, the higher the equilibrium price. In the extreme case of completely independent products ($d = 0$), the price corresponds to $p^B = \frac{\sigma + c'}{2}$ (since $\delta \rightarrow 0$ and $\beta = 1$ if $d \rightarrow 0$). Appendix 8.2 proves that this corresponds to the price p^M that a monopolist would charge. This makes sense, because if the two products are completely independent, then each of the duopolists is the monopoly supplier of a product that has no substitutes. If, instead, the two goods are perfect substitutes ($d = -1$), then it follows from equation (4) that $\delta = \beta$ and $\beta \rightarrow \infty$, and thus $p^B = \frac{\sigma}{\beta} + c' = c'$. In this extreme case, the duopoly price equals the marginal production costs and the duopolists do not earn any excess profits.

In the case of differentiated products and different production technologies, the firms' profits in equilibrium are:

$$\pi_i^B = \frac{\beta[2\beta\sigma_i - (2\beta^2 - \delta^2)c'_i + \delta(\sigma_2 + \beta c'_j)]^2}{[4\beta^2 - \delta^2]^2} - f_i \quad (8)$$

with $2\beta^2 - \delta^2 > 0$ since $\beta > \delta$. A firm's profits increase as its own marginal costs (c_i) and fixed costs (f_i) decrease. Therefore, spending on research and development (R&D) to reduce production costs (so-called process innovation) is a key business strategy in markets characterised by price competition. Furthermore, the profitability of enterprises is also determined by the degree of product differentiation. While prices for homogeneous products (perfect substitutes) decline to marginal costs and firm profits thus fall to zero, profits under maximum product differentiation

($\delta = 0 \Rightarrow \beta = 1$) are $\pi_i = \frac{(\sigma_i - c'_i)^2}{4} - f_i$ and thus identical to monopoly profits (see also Appendix 8.2).

In competition policy, the so-called **Lerner index** (Lerner, 1934) is often used as a measure of the market power of firms. This index measures by how much prices exceed marginal costs. In the case of symmetric demand ($\sigma_1 = \sigma_2 = \sigma$) and identical marginal costs ($c'_1 = c'_2 = c'$) we obtain:

$$L^B = \frac{p^B - c'}{p^B} = \frac{\sigma + (\delta - \beta)c'}{\sigma + \beta c} \quad (9)$$

If the products are perfect substitutes ($d = -1$, $\delta = \beta$ and $\beta \rightarrow \infty$), the Lerner index is zero. Otherwise, the Lerner index increases with the degree of product differentiation.

Case study 1: Pricing and product differentiation for yoghurt

In an empirical analysis, Loy and Weiss (2019) examined the influence of product differentiation on price formation on the German yoghurt market. The authors observed prices of 30 different yoghurt products in 432 stores over a period of 312 weeks and created a measure of product differentiation using various product characteristics (fat and calorie content, taste, brand name, etc.). Using these data, they estimated the effect on yoghurt prices of changes in raw milk prices (an essential component of the marginal cost of yoghurt production) and product differentiation. The authors show (a) that *ceteris paribus* the price of a product increases the more it differs from competing products, and (b) that yoghurt prices increase significantly over time with the price of raw milk. These results confirm two important predictions of the duopoly model: both marginal costs and the degree of product differentiation have a positive effect on price levels.

8.3.2. Competition with strategic substitutes (quantity competition)

While the price of a product was a firm's the decision variable (x_i) in the previous section, we now look at a situation in which the two competitors simultaneously choose the quantity produced. This situation is referred to

as **Cournot competition**.⁷ Using the demand function (equation 2), the first firm's profit maximisation problem is:

$$\max_{q_1} \pi_1 = (p_1 - c'_1)q_1 - f_1 = (s_1 - q_1 + dq_2 - c'_1)q_1 - f_1.$$

The first-order condition is:

$$\frac{\partial \pi_1}{\partial q_1} = s_1 - 2q_1 + dq_2 - c'_1 = 0.$$

Solving for q_1 gives the following best response function for the first firm:

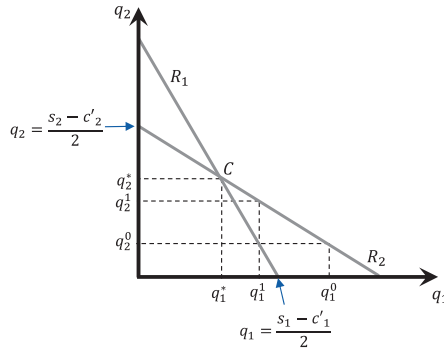
$$q_1 = \frac{s_1 + dq_2 - c'_1}{2} \equiv R_1. \quad (10)$$

If the products under consideration are substitutes ($d < 0$), then the reaction function in equation (10) shows that the best (i.e. profit-maximising) response by firm 1 to an increase in q_2 is a reduction in quantity q_1 : $\frac{\partial q_1}{\partial q_2} = \frac{d}{2} < 0$. An increase in q_2 will lower the market price, which is why firm 1 will reduce the quantity supplied. The fact that action and reaction offset shows that quantities are strategic substitutes (provided that the goods being considered are substitutes: $d < 0$).

Similarly, a response function for the second firm can be calculated as follows: $q_2 = \frac{s_2 + dq_1 - c'_2}{2} \equiv R_2$. A **Cournot-Nash equilibrium** is realised at the intersection of the two reaction curves: $q_i^C = \frac{2(s_i - c'_i) + d(s_j - c'_j)}{4 - d^2}$. In the simple case of symmetric demand ($s_1 = s_2 = s$) and identical marginal costs ($c'_1 = c'_2 = c'$), the two suppliers will supply the same quantity $q_1^C = q_2^C = q^C = \frac{s - c'}{2 - d} > 0$ for $s > c'$. This results in an equilibrium price of $p^C = \frac{s + (1 - d)c'}{2 - d}$, a profit of $\pi_i^C = \frac{(s - c')^2}{(2 - d)^2}$ and a Lerner index of $L^C = \frac{s - c'}{s + (1 - d)c'}$. The equilibrium in a symmetric Cournot duopoly is shown in Figure 8.2.

⁷ Antoine-Augustin Cournot (1801-1877) was a French mathematician and economist who first, within the framework of a formal model, described the strategic interaction of two competitors and the resulting equilibrium on the product market.

Figure 8.2: Reaction functions in a Cournot duopoly



Source: Own depiction.

We see from the expression for the equilibrium price that this price will increase if the willingness of consumers to pay or the quality of the product (s) increases, if the marginal costs of production (c') increase, and if the products become more differentiated and therefore less substitutable, i.e., if d increases ($\frac{\partial p^C}{\partial d} > 0$). If products are independent ($d = 0 \Rightarrow s = \sigma$), the price $p^C = \frac{s+c'}{2} = \frac{\sigma+c'}{2}$ corresponds to the monopoly price p^M , while the price is $p^C = \frac{s+2c'}{3}$ in the case of perfect substitutes ($d = -1$).

Comparing the results obtained for Bertrand and Cournot competition shows that the firms' prices and profits are lower under price competition: one might say that Bertrand competition is more aggressive than Cournot competition. This is particularly evident for homogeneous products. In this case (i.e. when the products are perfect substitutes), $d \rightarrow 1$. The Lerner index is then $L^B = 0$ for price competition, $L^C = \frac{s-c'}{s+2c'} > 0$ for quantity competition, and $L^M = \frac{s-c'}{s+c'}$ in a monopoly, with $L^B < L^C < L^M$.

8.4. Short-term strategic interaction in an oligopoly

The discussion of strategic interaction in a duopoly in the previous section is based on a number of simplifying assumptions (such as the quadratic utility function in equation (1) and the resulting linear demand functions in equations (2) and (3)). In this section we examine an oligopoly, i.e., a market with $n > 1$ identical firms, using a more general model. We assume that the

willingness of consumers to pay is given by the inverse demand function $p(Q)$, and that the i -th firm's variable and fixed costs of production are $c_i(q_i)$ and f_i , respectively. The profit maximisation problem for firm i is $\max_{q_i} \pi_i = p(Q)q_i - c_i(q_i) - f_i$, from which we obtain the following first-order condition: $\frac{\partial \pi_i}{\partial q_i} = \frac{\partial p(Q)}{\partial Q} \frac{\partial Q}{\partial q_i} q_i + p(Q) - c'_i(q_i) = 0$, where $c'_i(q_i) \equiv \frac{\partial c_i(q_i)}{\partial q_i}$ are the firm's marginal production costs. The term $\frac{\partial Q}{\partial q_i}$ captures the change in the total quantity supplied by all firms (Q) as a result of the change in the quantity supplied by firm i . If firm i supplies an additional unit of the good and all other firms respond by changing their combined supply by the amount θ , we can write $\frac{\partial Q}{\partial q_i} = 1 + \theta$. Parameter θ is called the conjectural variation. This parameter measures the aggregate response of all other firms to a change in quantity q_i : $\theta \equiv \frac{\partial Q_{-i}}{\partial q_i}$ (with $Q_{-i} = \sum_{j=1}^n q_j$ for $i \neq j$). As above, since all decisions are taken simultaneously, firm i does not know how its competitors will react if it changes q_i ; θ measures how firm i anticipates (i.e. conjectures) that its competitors will react. It follows from the first-order condition that:

$$p(Q) = -\frac{\partial p(Q)}{\partial Q} (1 + \theta) q_i + c'_i(q_i) \quad (11)$$

Expanding and rearranging terms for the case of n identical firms (i.e. for $c'_i(q_i) = c'(q)$ and $q_i = q = \frac{Q}{n}$) gives:

$$\begin{aligned} p(Q) &= -\frac{\partial p(Q)}{\partial Q} \frac{p(Q)}{p(Q)} \frac{Q}{Q} (1 + \theta) q + c'(q) \\ &= \frac{1}{\varepsilon} (1 + \theta) \frac{q}{Q} p(Q) + c'(q), \end{aligned} \quad (12)$$

where $\varepsilon = \left| \frac{\partial Q}{\partial p(Q)} \frac{p(Q)}{Q} \right|$ is the absolute value of the own-price elasticity of demand, which in turn depends on the extent of the products' substitutability. The resulting equilibrium price is:

$$p(Q) = \frac{c'(q)}{1 - (1 + \theta)/\varepsilon} = \mu c'(q) \quad (13)$$

The equilibrium price in an oligopoly is determined by marginal production costs ($c'(q)$), and the size of the **mark-up** $\mu = \frac{1}{1-(1+\theta)/\varepsilon n}$. The level of this mark-up, in turn, depends on three factors:

The own-price elasticity of demand (ε): The lower the absolute price elasticity of demand (i.e. the more urgently consumers want to buy the product), the higher the equilibrium price $p(Q)$. The price elasticity of demand also highlights the importance of product differentiation: the more readily consumers substitute the products for one another, the more price-elastic demand will be (from $d \rightarrow -1$ it follows from equation (4) that $\beta \rightarrow \infty$ and thus $\varepsilon \rightarrow \infty$), and the lower the equilibrium price.

The number of suppliers (n): The higher the number of competitors, the lower the equilibrium price.

The behaviour of the suppliers (θ): The concept of conjectural variation is the main innovation of this oligopoly model compared with the duopoly model described above. For this reason, we study its implications in greater detail in the following.

Very aggressive market conduct is reflected in a conjectural variation of $\theta = -1$. For example, if firm i reduces output in an attempt to raise the market price, $\theta = -1$ implies that competitors will completely counteract this measure by increasing their output by the same amount. As can be seen in equation (13), if $\theta = -1$ the mark-up μ equals one and the price equals marginal cost. Note that marginal cost prices arise whenever $\theta = -1$, regardless of the number of competitors (as long as $n > 1$) and the price elasticity of demand. This illustrates an important insight: the number of firms is not, *per se*, a reliable indicator of the intensity of competition in a market. As we saw above, two competitors can be enough to guarantee perfect competition and thus efficient market outcomes. Hence, focusing exclusively on indicators of market structure (i.e. the number of firms and their relative sizes) can be misleading; competition policy must focus on firms' actual conduct.

Perfect competition ($p = c'(q)$) will also be the result of an atomistic market, i.e. if $n \rightarrow \infty$. Furthermore, the oligopoly price equation (13) above can also depict monopoly outcomes as a special case. A conjectural variation of $\theta = n - 1$ means that the firms' conduct is perfectly synchronised; an increase (a reduction) in q_i by one unit is copied by all other firms; each increases (decreases) quantity q_j by one unit and,

therefore, quantity Q_{-i} increases (decreases) by another $n - 1$ units. This synchronised behaviour leads to the same market outcome as if the n firms had reached an explicit collusive agreement, and the market price is exactly the same as in a monopoly: $p = \frac{c'(q)}{1+1/\varepsilon} = p^M$.

Finally, a conjectural variation of $\theta = 0$ is referred to as Cournot-Nash behaviour. In this case, firm i assumes that its competitors will not react at all if it changes its output.

The concept of conjectural variation is particularly useful for illustrating the different paradigms of competition policy. The Lerner index of firm i 's market power can be calculated from equation (6) as follows: $L_i^{CV} = \frac{p(Q) - c'(q_i)}{p(Q)} = \frac{1}{\varepsilon} (1 + \theta) \frac{q_i}{Q} = \frac{1}{\varepsilon} (1 + \theta) r_i$, where $0 < r_i = q_i/Q \leq 1$ is the market share of supplier i . The aggregate Lerner index for the entire market, can be calculated from the individual firms' Lerner indices, each weighted by the respective firm's market share:

$$L^{CV} = \sum_{i=1}^n r_i L_i^{CV} = \sum_{i=1}^n \frac{1}{\varepsilon} (1 + \theta) r_i^2 = \frac{1 + \theta}{\varepsilon} \sum_{i=1}^n r_i^2 = \frac{1 + \theta}{\varepsilon} HHI \quad (14)$$

where $HHI = \sum_{i=1}^n r_i^2$ is the Herfindahl-Hirschman index of supplier concentration. This index ranges from $1/n$ (when all firms have an equal market share of $r_i = r = 1/n$) to 1 (in the case of a monopoly, when $r_i = 1$ and $n = 1$).

Equation (14) can be interpreted in a number of ways. First, for a given consumer behaviour (represented by the price elasticity of demand ε), the mark-up on marginal costs increases with market concentration (measured by the Herfindahl-Hirschman index HHI). Thus, competition policy should aim at preventing (excessive) increases in market concentration, by regulating mergers or by breaking firms that have market power. The so-called Harvard School of competition policy supports this view. According to the Chicago School, however, a high Lerner-index is not (necessarily) caused by firms exploiting their market power in concentrated markets. Instead, the superior efficiency (i.e. low marginal costs) of individual firms leads to both a high Herfindahl-Hirschman index and a high Lerner-index. In this case, consumers benefit when inefficient firms are taken over by efficient competitors, and restrictive competition policies that prohibit such takeover will result in welfare losses. Divergent perspectives and sometimes

diametrically opposed competition policy recommendations by economists are often a consequence of different interpretations of equation (14).

The key advantage of the conjectural variation model is that the entire range of competitive conduct (from perfect competition to monopoly) is represented by a single parameter (θ). This makes it easy to analyse the consequences of entrepreneurial behaviour for market outcomes such as prices, quantities, and profits in equilibrium. However, and this is the main drawback of the model, entrepreneurial behaviour is only represented by θ and not explained; θ is exogenous. Why and under what conditions businesses act aggressively or collude remains unanswered.

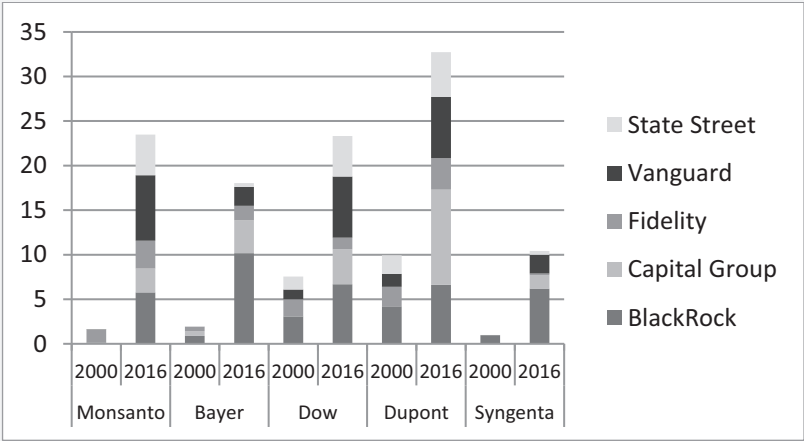
In addition, as mentioned above, this static model does not capture reactions in the usual sense of the word. Firms do not react to previous actions by other firms in the model, instead they anticipate the reactions of other firms when they make simultaneous decisions. Hence, long-term (strategic) business decisions are, by definition, not considered.

Case study 2: (Common) ownership matters

The basic oligopoly models discussed above ignore important issues, such as the internal organization of firms and their ownership. Ownership and market power, however, are interrelated. The rapid growth in investment funds has had a significant impact on the ownership structure of large firms. Institutional investors today hold substantial shares in multiple firms in the same industry (common ownership). Common ownership of same-industry firms incentivizes both investors and managers to maximize portfolio rather than firm profits, which could lead to anti-competitive outcomes.

Torshizi and Clapp (2021) study the impact of common ownership in the US seed sector. In this sector, the 'Big 6' (Monsanto, Bayer, Dow, DuPont, Syngenta, and BASF) control the majority of seed and chemical sales. The authors document that the largest asset management companies (BlackRock, Vanguard, State Street, Capital Group, and Fidelity) collectively hold substantial shares in the 'Big 6' (see Figure 8.3) and argue that this network of common owners and the ownership linkages among rival firms could further reduce the intensity of competition in this sector.

Figure 8.3: Total percentage ownership shares of institutional investors in Bayer, Monsanto, DuPont, Syngenta, and Dow, 2000–2016.



Source: Torshizi and Clapp (2021, p. 42).

Following previous research in the area of finance and industrial organization, Torshizi and Clapp (2021) show how common ownership moves the market from the more competitive standard Cournot-Nash oligopoly equilibrium toward a less competitive outcome (which implies an inward shift of firms’ reaction functions – more details are provided in an online appendix of their paper). This implies an increase in “effective concentration”, which can be represented by a modified Herfindahl-Hirschmann Index $MHHI = HHI + \Delta$ (where Δ measures the effects of common ownership). The authors’ empirical calculations for the US seed market show that the mean of Δ across corn, soybeans and cotton is almost as high as the HHI market concentration index. In an econometric analysis, they investigate the impact of common ownership on seed prices for soybean, corn, and cotton seed prices over the 1997–2017 period. Their analysis suggests that, on average, 14.6% of the increase in corn, cotton, and soybean seed prices can be attributed to the increase in common ownership. Common ownership matters in the US seed market.

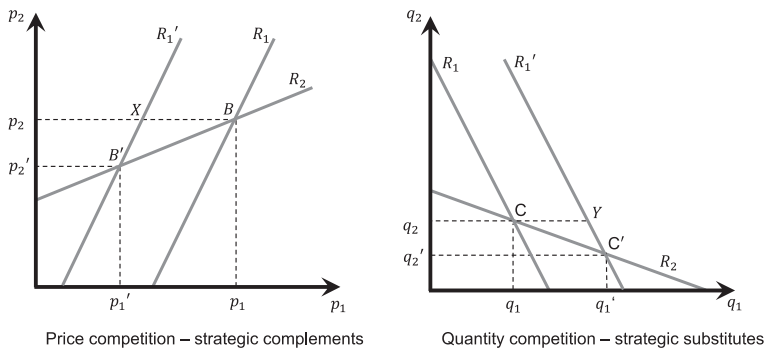
8.5. The behaviour of businesses in the long run

If a model only depicts a single point in time, it cannot distinguish between tactical (short-term) and strategic (long-term) decisions. Such a model cannot provide answers to many important questions. For example: What strategies can firms use to force existing competitors out of the market (or prevent new competitors from entering)? Under what conditions are mergers between firms profitable? How much should an entrant differentiate its product from that of an incumbent, and should the entrant supply higher or lower quality? How much should a firm invest in advertising or in R&D to improve production technology? These and many other strategic decisions change the environment (market structure) within which firms make the short-term (tactical) decisions discussed in previous sections (i.e. decisions on optimal prices or quantities).

Strategic questions are analysed in modern industrial economics using two- or multi-stage games. These games show that the decision for or against a particular strategic measure is influenced by the nature of competition (strategic complements or substitutes). In the following we will illustrate this using a simple example of a market with two firms (duopoly) and two time-periods.

Consider the first firm's strategic decision to invest in improving its production technology. From the firm's point of view, expenditures on research and development (R&D) in the first period (t_1) constitute fixed costs. If the R&D efforts are successful, firm 1 will be able to compete with firm 2 in the following period (t_2) with a more efficient production technology (i.e. the first firm's marginal costs will decrease). This competitive advantage will influence the firm's tactical price or quantity decisions in the second period. When making a strategic decision, firm 1 must consider how firm 2 will respond to this decision. The two diagrams in Figure 8.4 show that the second firm's reaction in period t_2 will depend on the nature of the competition, and that this plays an essential role in assessing the profitability of the original strategic R&D decision in t_1 .

Figure 8.4: Direct and indirect effects of R&D expenditure



Source: Own depiction.

Figure 8.4 shows the tactical decisions of the two firms in t_2 . Due to the first firm's investments in R&D in t_1 , its marginal costs decline. This reduction in marginal costs changes the first firm's choice of an optimal price or quantity in t_2 . In the case of Bertrand competition (price competition), reduced marginal costs will lead firm 1 to lower the price of its own product. This tactical decision is apparent from equation (5), in which $\frac{\partial p_1}{\partial c_1'} = \frac{1}{2} > 0$: for a given competitor's price (p_2), if c_1' decreases then so does the optimal price p_1 . This corresponds to a shift of the first firm's reaction function from R_1 to R_1' in the left panel of Figure 8.4. Firm 2 will respond to a decline in p_1 with a price reduction for its own product. The adjustments in both prices in period t_2 as a consequence of the first firm's R&D expenditures in t_1 (and the associated reduction in marginal costs) are represented by a movement from point B to a new equilibrium point B' in Figure 8.4.

For the overall assessment of the first firm's strategic decision, two countervailing effects are important. First, the first firm's profit increases due to the reduction in its production costs and the associated optimal adjustment of price p_1 . This direct effect corresponds to the movement from point B to point X (with unchanged p_2) in the left panel of Figure 8.4. Second, however, the competitor's response (the indirect or strategic effect) acts in the opposite direction; firm 2 reduces its price to p_2' , which reduces the demand for firm 1 and thus reduces its profits. This strategic effect of the competitor's reaction corresponds to the movement from point X to point B' in Figure 8.4.

Hence, under Bertrand competition (strategic complements) the direct and indirect effects of the first firm's strategic decision to invest in R&D work in opposite directions. In contrast, the same strategic decision under conditions of Cournot competition (strategic substitutes) will trigger mutually reinforcing direct and indirect effects. The right panel of Figure 8.4 shows that in the case of Cournot competition the reduction in marginal costs will lead to an increase in the first firm's output; the reaction function shifts to the right from R_1 to R_1' . This shift is due to the fact that $\frac{\partial q_1}{\partial c_1} = -\frac{1}{2} < 0$ in equation (10). The direct effect therefore leads to a move from point C to point Y . The second firm's response benefits firm 1 in this case; firm 2 will, in its own interest, reduce its output from q_2 to q_2' , to which firm 1 will respond by further increasing its output to q_1' . The indirect (strategic) effect therefore leads to a movement from point Y to point C' . Both the direct and indirect effects increase profits for firm 1.

This example illustrates that the profitability of a firm's long-term (strategic) decisions is determined by the nature of market competition (strategic complements or strategic substitutes).⁸ The form of competition in a market thus not only influences the level of prices or quantities in the short term but is also decisive for assessing the profitability of long-term business strategies. The effects described in the example above are relevant not only for the specific case of R&D expenditures, but also for other strategic decisions such as advertising, the outsourcing of production processes, the bundling and coupling of products, and strategic trade policy.

8.6. Extensions and modifications

8.6.1. The choice between price and quantity competition

Which strategy (price- or quantity-setting behaviour) should a firm choose? Under what conditions would it be profitable for a firm to fix a particular price and let consumers decide how much they want to consume (Bertrand competition)? When would the alternative strategy of setting the optimal quantity (Cournot competition) be more profitable? What happens when firms choose different strategies? While we have so far assumed that the

⁸ Fudenberg and Tirole (1984) also distinguish between "soft" and "tough" strategies. In the situation just described, reductions in marginal costs caused by expenditures on R&D will induce firm 1 to lower its price and increase output. This will reduce the competitor's profits. Such strategies are called "tough". A "soft" strategy increases the competitor's profits.

form of competition (price or quantity competition) is exogenous, we now want to examine the choice between these two strategies as a managerial decision.

When firms are not perfectly informed about consumer demand, the choice between price and quantity competition depends crucially on the characteristics of the production technology. If production technology is characterised by sharply increasing marginal costs (i.e. an increase in output is associated with high additional costs in the short term), quantity setting is more profitable. If firms were to fix prices instead of quantities in this situation, any fluctuation in demand would have to be met by adjusting quantities, which would entail high additional costs. However, if marginal costs are constant (or only rise moderately), price-setting behaviour becomes more attractive. Sharply increasing marginal costs are to be expected *ceteris paribus* where firms face short-term capacity constraints. In contrast, constant or moderately increasing marginal costs are more plausible where firms have excess capacity or are taking a long-term perspective.

What happens if individual businesses choose different strategies? Assume for example, that one firm opts for the Bertrand strategy, while the competitor chooses a Cournot strategy. In this case, which is an example of a so-called **mixed oligopoly**, it can be shown that Cournot behaviour will prevail over Bertrand behaviour as a dominant strategy. Overall, the assumption of quantity-setting behaviour has established itself as the starting point for numerous applied analyses, especially in studies of relatively homogeneous products. As Shapiro (1989, p. 346) concludes in an excellent survey: "For homogenous goods, Cournot's model remains the workhorse oligopoly theory".

8.6.2. Competition with endogenous market structure

So far, we have analysed imperfect competition under the assumption that the market structure (i.e. the number of competitors, production technology and consumer preferences) is exogenous. In this framework, the given, exogenous market structure determines firm conduct (the aggressiveness of competition as expressed in the choice of prices and quantities, advertising expenditure, investments in R&D, etc.) which again affects market performance in equilibrium (prices, quantities, profits and consumer welfare). However, this framework, which is referred to as the classic structure-conduct-performance (SCP) approach in industrial economics, ignores the fact that some firms might exit the market if they perform

poorly, or that new firms might be attracted by high profits and enter the market. Market structure (the number of firms, for example) therefore depends on firms' conduct and performance, and is thus endogenous. Similarly, attempts by firms to improve their production technology (through R&D) or to influence consumer preferences (e.g. through advertising) might change production technology and consumer demand; effects which are also not considered in the SCP approach.

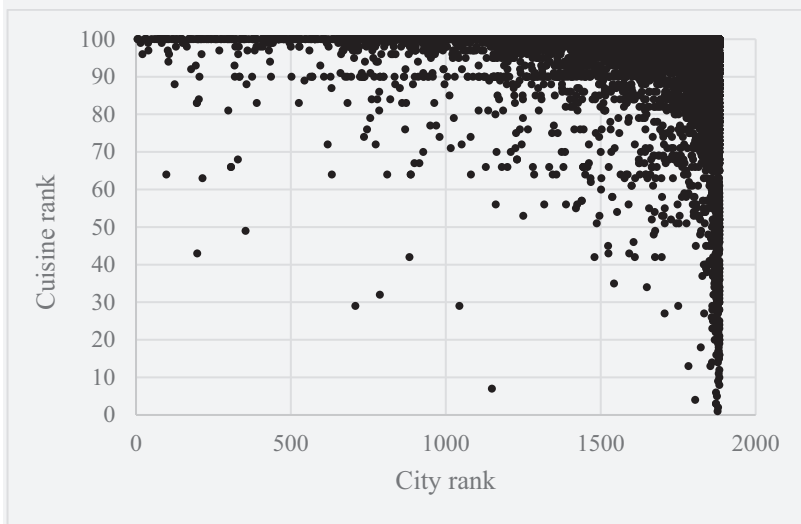
Modern theories therefore use two- or multi-stage models to account for the endogeneity of market structure. In a first stage, the long-term (strategic) decisions (the number of competitors or products supplied in the market, the amount of expenditure on advertising or R&D, etc.) are modelled. On this basis, the short-term (tactical) decisions (choice of prices or quantities) are investigated in a second stage. We use backward induction to identify the sequence of actions in the Nash equilibrium. This means that we first determine the optimal second-stage (short-term, tactical) decision for each possible outcome of the first-stage (long-term, strategic) decision. Given the optimal decisions from the second stage, we then calculate optimal strategic decisions in the first stage. The following case study illustrates this approach by investigating market structure (the number of restaurants and cuisines) in the food sector.

Case study 3: Competition between restaurants

Over the past decades, out-of-home consumption has become a major economic factor in developed economy food sectors. In a two-stage, spatial model, Schiff (2015) describes the behaviour of firms in this business and their customers. Firms have to make two decisions: the first-stage, long-term (strategic) decision is whether to open a new restaurant in a city and, if so, what type (cuisine) of restaurant. Given the long-term decisions of all competitors, and the potential customers' behaviour, short-term price decisions are made in the second stage. Under the assumption that restaurants enter a market only if expected profits are positive (or that they leave a market if profits are negative), Schiff calculates so-called "entry thresholds". These indicate the number of consumers in a given region required for profitable market entry. The larger the number of potential customers in a region (or the higher their purchasing power), the greater the number of restaurants in that region. The entry thresholds depend, among other things, on the preferences of consumers for particular cuisines. While a restaurant that serves a standard cuisine for a broad public can operate profitably in a comparatively small town, a restaurant that serves a specialty cuisine for select customers can only survive in a comparatively large city.

Replicating Schiff's (2015) analysis, Pennerstorfer et al. (2020) apply this model to observations of 101 cuisines in 24,454 restaurants in 1,883 Austrian municipalities. In Figure 8.5, each point represents a city/cuisine combination. On the horizontal axis the cities are sorted according to their rank in the number of different cuisines offered ("city rank"). The city with the smallest number of different cuisines is ranked first (to the left of Figure 8.5). On the vertical axis ("cuisine rank") the 90 cuisines are sorted according to their prevalence in the 1,883 municipalities. The cuisine which is offered in the fewest municipalities is ranked first (to the bottom of Figure 8.5). We see that cities with many restaurants (to the far right of the figure) also have a larger variety of different cuisines, including cuisines that are offered only rarely (to the bottom of the figure). Cities with few restaurants (on the left), however, only offer common or widespread cuisines (at the top of the figure) that are offered in most municipalities.

Figure 8.5: Relationship between 'city rank' and 'cuisine rank' (explanation in text)



Source: Pennerstorfer et al. (2020, p. 48).

8.6.3. Competition over time

A limitation of static models of imperfect competition is that they neglect time. All decisions are taken simultaneously; there is neither a future (for which one might be willing to make investments today) nor a past (from which economic agents could learn for today's and tomorrow's decisions). Game-theoretical

models show that the behaviour of economic agents depends heavily on whether they interact only once or whether they meet and interact repeatedly. In the latter case, in **repeated games**, it is important to distinguish between finite games in which players are aware that the game will be played a known, finite number of rounds, and games in which the players do not know how many rounds will be played (so-called **super games**). Super games deliver important new insights, particularly regarding the likelihood of collusion between firms. While static models and finite games cannot adequately explain collusion between firms, super games generate a number of empirically supported hypotheses about the occurrence and durability of collusive behaviour and cartels. Without going into detail, the following ten stylised facts summarise the main results in the literature (see Feuerstein 2005):

- i. A small number of suppliers and high barriers to entry facilitate collusive behaviour.
- ii. Large differences in market shares between individual firms make collusive agreements more difficult.
- iii. Frequent interactions among firms and a high frequency of price adjustment facilitate collusive behaviour.
- iv. High transparency in markets facilitates the maintenance of collusive agreements.
- v. For a given number of firms, collusive agreements are more likely to be maintained in growing markets.
- vi. Fluctuations in demand (e.g. pronounced seasonal fluctuations) reduce the likelihood of collusion.
- vii. Innovations (product and process innovations) make it difficult to maintain agreements.
- viii. Large differences in cost structures or capacity limits between firms make collusive behaviour more difficult.
- ix. Multi-market contacts (i.e. situations in which firms with broad product palettes compete with one another on different markets) facilitate collusive behaviour.
- x. A high price elasticity of demand reduces the likelihood of collusion and cartels between firms.

An extensive empirical literature analyses food industry cartels such as the Lysine Cartel and the Citric Acid Cartel (see Case Study 4). Levenstein and Suslow (2006) provide a detailed overview of the factors that influence the success and longevity of cartels.

Case study 4: The Lysine Cartel and other examples from the food industry

In the mid-1990s, five global suppliers of lysine, an amino acid used in animal production, started price-fixing. The companies involved were American (ADM), Japanese (Ajinomoto and Kyowa Hakko Kogyo), and Korean (Sewon America Inc. and Cheil Jedang Ltd.). A senior ADM official, Mark Whitacre, made tape recordings of conversations between the cartel members for the FBI. The US Bureau of Competition imposed heavy fines on ADM, and responsible company managers received prison sentences. The story of the lysine cartel was made into a film "The Informant" starring Matt Damon. A number of economic studies (including Bolotova et al. 2008) have analysed the lysine cartel in great detail.

The food sector provides numerous further examples of collusion and cartels. These include the following cases in Germany alone: the coffee market (Bundeskartellamt 2009); the market for milk and dairy products (Bundeswettbewerbsbehörde 2013b); the beer market (Bundeskartellamt 2014a); sausage production (Bundeskartellamt 2014b); sugar production (Bundeskartellamt 2014c); and food retailing (including the Bundeswettbewerbsbehörde 2013a; Bundeskartellamt 2016).

8.6.4. Competition in space

Consumers are diverse; they have different preferences and face different budgetary constraints. We have ignored this consumer heterogeneity in the model above by assuming a representative consumer (sections 8.2 and 8.4). This model does account for product variety, and a preference for variety is built into the representative consumer's utility function.⁹ However, since this preference for variety is represented by a single parameter, the model

⁹ In the utility function in equation (1), the marginal utility of good i decreases with the quantity consumed of that good (q_i), and increases with the quantity consumed of the substitute (q_j). Hence, a balanced consumption bundle (in which the available budget is divided equally among all n goods) generates a higher utility than a specialised bundle (in which the entire budget is spent on only one product only).

assumes that all products are equally substitutable: the products are different but they are all equally different from one another (see footnote 2), and each product therefore competes with equal intensity with each other product (global competition). In reality, of course, some products are close substitutes in the eyes of consumers and therefore in close competition with one another, while others are so different that they hardly compete.

Products can be differentiated geographically as well as in terms of characteristics such as ingredients, taste, quality, or packaging. Geographical differentiation occurs when transport costs are so high that firms in a geographic region compete with one another, but not with firms in other regions. The following case study illustrates differentiation based on product characteristics for the cereals market.

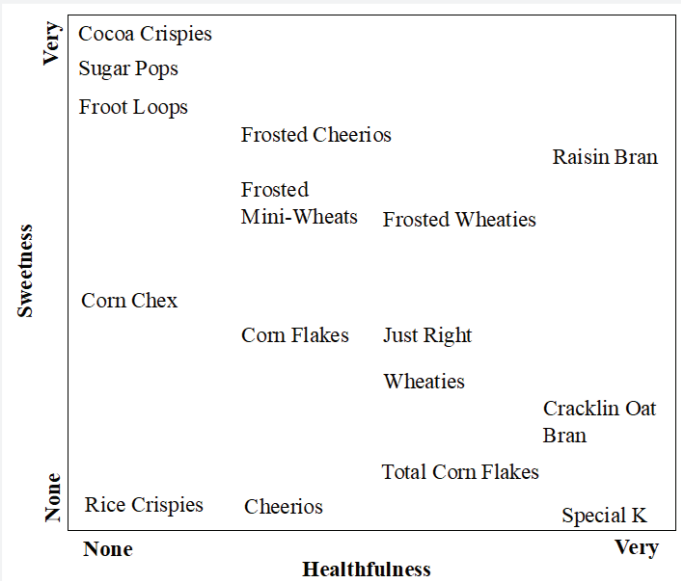
Case study 5: Product differentiation in the breakfast cereal market

In the business literature, so-called perceptual maps are often used to position individual products or services in characteristic space (Kotler and Keller, 2012). The example in Figure 8.6 depicts competition between individual brands in the US breakfast cereal market.

In Figure 8.6, various cereal brands are positioned in a two-dimensional space based on the characteristics Healthfulness and Sweetness. The perceptual map suggests, for example, that Special K is a close substitute for Total Corn Flakes, so that these products compete intensively for customers who value Healthfulness and prefer cereals that rank low in Sweetness. In contrast, the large distance between Special K and Cocoa Crispies suggests that competition between these products for consumers is weak.

Models of product differentiation account for the fact that consumers differ because they are located in different places and have different preferences for product characteristics. Products and consumers can be positioned in space based on these differences, where ‘space’ can refer to location but also characteristic space, as for example in the perceptual map above. Competition on goods markets takes place only between firms that are located close to one another in ‘space’. If this is the case, the demand for one product is strongly influenced by the prices of similar (or nearby) products, but only reacts weakly to the prices of other products.

Figure 8.6: Perceptual map of the US breakfast cereal market



Source: Pindyck (2011).

8.7. Empirical analyses of market power

“Market power is like the wind. You can feel it but you cannot see it”.
Kohls and Uhl (2002, p. 270).

8.7.1. Structure-conduct-performance approach

(How) can we prove the existence of market power empirically? The industrial organisation literature has applied various methods. As described above, the traditional SCP approach assumes that market structure (e.g. the number of competitors) is exogenous. Market structure determines firm conduct, which in turn determines market outcomes (e.g. equilibrium prices and profits). For example, equation (13) shows that equilibrium prices are determined both by market structure (the number of suppliers n , the marginal production costs c' , and the preferences of consumers expressed by the price elasticity of demand ε) as well as by firm conduct (the intensity of competition represented by the conjectural variation θ). Empirical studies based on the SCP approach typically examine the relationship between indicators of market structure and market performance in order to

draw conclusions about the (unobservable) conduct of competitors in between. The following case study illustrates this procedure.

Case study 6: Market power in the US food industry

Empirical SCP analyses attempt to establish a direct relationship between market structure (measured for example by supplier concentration) and market performance (e.g. the level of profits). Zellner (1989) is a typical example of SCP analysis. Based on cross-sectional data for 75 branches of the US food industry, Zellner analyses how structural characteristics of these industries influence firm behaviour (such as expenditure on advertising) and performance (profitability). The following equation shows one of the many empirical results provided in this study (Zellner 1989, p. 113, t-statistics of the estimated coefficients in brackets):

$$\begin{aligned} \text{Lerner index} = & -1.76 + 14.82\text{Growth} + 0.40\text{CR4} + 1.22\text{ASR} + 0.13\text{Brands} \\ & (0.31) \quad (2.82) \quad (3.96) \quad (1.72) \quad (2.49) \\ & -1.58\text{Innov} + 0.16\text{COR} - 0.99\text{Disp} \\ & (1.85) \quad (1.91) \quad (2.67) \end{aligned}$$

Zellner calculates the Lerner index (equation 9) using average production costs. He observes a significant positive effect of industry growth rates (*Growth*), the concentration ratio (*CR4*, the share of the four largest firms in total industry turnover), the capital-output ratio (*COR*), the advertising-to-sales ratio (*ASR*), and product differentiation measured by the number of different products (*Brands*). He also notes a significant negative effect of innovation (*Innov* measures the number of new products in an industry) and an indicator of geographical dispersion (*Disp*) on the Lerner index.

The analysis presented in case study 6 is based on a cross-sectional regression over a large number of different branches of the food industry. One weakness of such analyses is that they assume that market structure has the same effect on performance in all branches of an industry. In addition, while these analyses might find that concentration and profitability are correlated, the reasons underlying such correlations remain unclear. A positive correlation might, for example, be the consequence of firms exercising market power in highly concentrated branches or industries (Harvard school). However, it might also be due to the higher efficiency of larger companies, which translates into both higher profits and higher concentration ratios (Chicago school). These weaknesses of the traditional

SCP approach can be mitigated by analysing prices (rather than profits) with firm-level data from a specific industry or branch, as illustrated in the following example.

Case study 7: Market power in food retailing in Vermont

Based on price observations for 121 representative products in August 1981, Cotterill (1986) calculated a price index for individual supermarkets in 18 different regions of Vermont in the US. He then looks for evidence that differences in price levels between supermarkets are caused by regional differences in supermarket concentration, while controlling for other characteristics of the individual supermarkets. The following equation shows the results of one of his estimated models (Cotterill 1986, p. 383, t-statistics in brackets).

$$PI = 99.837 + 8.912HHI + 2.241Indep + 0.007Size + 4.665Sales/Size + 0.002Dist \\ (5.49) \qquad (2.72) \qquad (0.82) \qquad (2.10) \qquad (0.33) \\ -0.017Popgr + 0.044Income \\ (0.36) \qquad (0.17)$$

The food price index (*PI*) increases significantly with the Herfindahl-Hirschman-Index of regional supermarket concentration (*HHI*). In addition, *PI* is significantly higher for independent supermarkets (*Indep* = 1) and supermarkets with high sales per unit area (*Sales/Size*). Supermarket size (*Size*, measured by their sales area), the regional population growth rate (*Popgr*), and average per capita income in the region (*Income*) have no significant influence on *PI*. The author concludes: "The estimated structure-price relationships suggest that firms with strong market positions enjoy high profits because they raise prices rather than lower unit costs" (p. 385). Based on a much larger number of observations, Asplund and Friberg (2002) as well as Barros et al. (2006) obtain similar results for supermarkets in Sweden and Portugal respectively.

Caves (2007) outlines the merits of the traditional SCP approach compared with more recent empirical approaches. An important advantage of the traditional SCP approach is that it can be used to analyse the intensity of competition and its effects across industries, while more recent studies often focus on firms in a single industry. However, this advantage is also a weakness, because industries differ in many respects besides the intensity of competition, and it is difficult to adequately control for all of these other

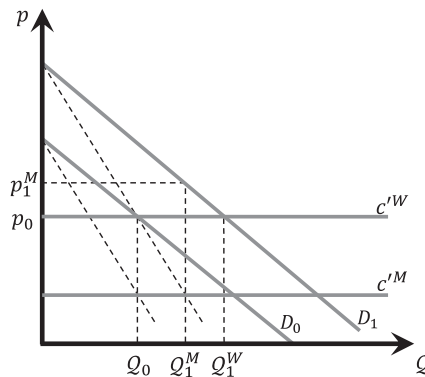
possibly confounding factors. Furthermore, as mentioned above, the SCP approach assumes that market structures are exogenous and therefore ignores the fact that firms that perform poorly will exit the market, and other firms might enter the market if industry profits are high. It also fails to account for attempts by firms to improve their production technologies (process innovation) and to influence consumer preferences (e.g. through advertising and/or product innovation). Finally, the SCP approach does not analyse the actual behaviour or conduct of firms. It focusses on the relationship between market structure and market performance, while conduct remains unobserved and implicit between them. Hence, compared with more recent approaches, some of which will be discussed below, the SCP only estimates a reduced form of a more extensive structural model.¹⁰

8.7.2. Direct measurement of the Lerner index

Measuring market power would be straightforward if, in addition to product prices, firms' marginal production costs were also observable. In this case, it would be straightforward to calculate the Lerner index ($L_i = \frac{p_i - c_i^l}{p_i}$), which provides explicit information on a firm's mark-up. However, reliable information on a firm's marginal costs is rarely available. Empirical studies often solve this problem pragmatically by using average costs to approximate marginal costs (see for example case study 6). This, however, can lead to systematic bias, because marginal and average costs are only identical if marginal costs are constant and fixed costs equal zero. In addition, and perhaps more important, firms with market power have strong incentives to misreport their marginal costs. Figure 8.7 illustrates this problem for the simple case of constant marginal costs.

¹⁰ Reiss and Wolak (2007) discuss in detail the distinction between estimating a model in 'structural form' and in 'reduced form', and the advantages and disadvantages of the two approaches.

Figure 8.7: Identifying market power: the case of demand shocks



Source: Own depiction.

Consider a market with demand curve D_0 on which we observe the price p_0 and aggregate quantity Q_0 . This price-quantity combination could be the outcome of a monopoly market with marginal cost curve c^M . However, the identical price-quantity combination could also result from perfect competition if the marginal cost curve is c^W . A monopolist will have a strong incentive to inflate its actual marginal costs in an attempt to hide the fact that it is exercising market power from the competition authorities. Philips (1996) calls this the **indistinguishability theorem** and formulates "10 commandments" for firms with market power in their dealings with competition authorities. The fifth commandment is "Thou shalt exaggerate the level of costs" (p. 509). In addition to eight further recommendations for misleading competition authorities, the tenth commandment is: "Thou shalt otherwise tell the truth" (p. 510).

Despite this problem, it is sometimes possible to identify market power even when marginal costs are unknown by observing how firms react to exogenous shocks. Consider, for example, a firm's optimal response to an exogenous demand shock. If the demand curve shifts outwards to D_1 in Figure 8.7, prices will not react in a competitive market (with constant marginal costs of c^W); we will only observe an increase in the equilibrium quantity from Q_0 to Q_1^W . In contrast, a monopolist with constant marginal costs of c^M would increase prices to p_1^M in response to the same shock. The situation described in Figure 8.7 can be extended to the case of rising rather than constant marginal costs (Breshnahan 1982) and to differentiated

products (Nevo 1998). In this way it is possible to use market reactions to demand fluctuations to draw conclusions about market structure.

8.7.3. Analyses of price transmission

Even if marginal costs are unobservable, it is frequently possible to observe individual components of these costs. The prices of dairy products, for example, will depend critically on the producer price of milk (see case study 1). The producer price of milk is not the only component of a dairy's marginal costs of yoghurt, of course, but it is a large and essential component. Under very specific assumptions about the form of demand and cost functions, the transmission of exogenous cost shocks also would allow identification of market power. In Chapter 9 we analyse vertical price transmission in the food chain, in other words the question how prices and margins in the food chain react to changes in the prices of agricultural raw products such as milk, wheat and slaughter pigs. Among other things, we demonstrate in Chapter 9 that market power can be identified using empirical price transmission analysis, but only under exacting conditions that are often not met.

8.7.4. Production-theoretic approaches

Following Hall's (1988) seminal study, production-theoretic approaches attempt to measure marginal costs (c') and the mark-ups ($\mu = \frac{p}{c'}$) by econometric estimation of production or cost functions. If we consider the simplest case, producing the amount q requires only two inputs, labour l (which is variable) and capital k (which is fixed). Production $q = q(l, k)$ is associated with the cost function $c = wl + rk$ where w and r are the prices of l and k , respectively. For a given output $\bar{q}$, a firm's cost minimisation problem can be stated in the following Lagrange function: $\min L = wl + rk + \lambda[\bar{q} - q(l, k)]$. The Lagrange multiplier $\lambda = \frac{\partial L}{\partial \bar{q}}$ shows how production costs change with the output quantity $\bar{q}$, and thus is a measure of marginal costs ($\lambda = c'$). The first-order condition for the variable factor (l) is: $\frac{\partial L}{\partial l} = w - \lambda \frac{\partial q(l, k)}{\partial l} = 0$. After expanding by $\frac{l}{q(l, k)}$ and transforming the equation, we obtain: $\frac{w}{\lambda} \frac{l}{q(l, k)} = \frac{\partial q(l, k)}{\partial l} \frac{l}{q(l, k)}$. Combining this expression with the definition of the mark-up $\mu = \frac{p}{c'} = \frac{p}{\lambda}$ leads to $\mu = \left(\frac{\partial q(l, k)}{\partial l} \frac{l}{q(l, k)} \right) / \left(\frac{w}{p} \frac{l}{q(l, k)} \right) = \frac{\xi}{\frac{wl}{pq(l, k)}}$, where $\xi \equiv \left(\frac{\partial q(l, k)}{\partial l} \frac{l}{q(l, k)} \right)$ is the

production elasticity of the variable factor l , and $\frac{wl}{pq(l,k)}$ is the cost share of input l in the value of production. The production elasticity ξ can be determined by estimating a production function or a cost function.¹¹ The ratio between the estimated production elasticity and the cost share of l is an estimate of the mark-up ($\mu = \frac{p}{c_l}$), and also provides us with an estimate of the Lerner index (since $L = \frac{\mu-1}{\mu}$, see equation 9).

Production theoretic approaches are particularly well-suited for the analysis of markets when the production technology can be represented econometrically (i.e. if suitable data and good knowledge of production technology are available). As the following case study shows, this approach can take specific production conditions, such as economies of scale, into account. Economies of scale will lead to systematic differences in mark-ups between large and small firms.

Case study 8: Market power on the German beer market

The German beer brewing companies have repeatedly been the subject of antitrust proceedings by the German competition authorities (Bundeskartellamt 2014a and 2016). Based on data from 197 breweries over 13 years, Karagiannis et al. (2018) estimate a flexible form of production function (a so-called translog production function) and calculate production elasticities for three inputs (labour, intermediate inputs, and capital). The authors calculate an average mark-up of 1.65 from the ratio of the production elasticity of intermediate inputs to their cost share; this corresponds to a Lerner index of $L = 0.4$. The results of the econometric estimation also point to the existence of economies of scale in the brewing industry. Economies of scale mean that large firms can charge higher mark-ups than small firms. The authors' calculations suggest a Lerner index of $L = 0.47$ for large breweries (with an annual production of more than 100,000 hl) compared with $L = 0.28$ for small breweries (with an annual production of less than 25,000 hl).

¹¹ For example, given the Cobb-Douglas production function $q = \tau l^\xi k^\gamma$, the production elasticity ξ can be estimated by taking logarithms, which gives $\ln(q) = \ln(\tau) + \xi \ln(l) + \gamma \ln(k)$ (see Appendix 2.3 in chapter 2). Many empirical studies estimate cost functions (the dual representation) rather than production functions to obtain production elasticities.

8.7.5. Market power and consumer demand for differentiated products

As shown in section 8.4., the Lerner index is calculated from a conjectural variation model as follows (see equation 14): $L = \frac{p-c'}{p} = \frac{1+\theta}{\varepsilon} HHI$. Direct measurement of L requires information on prices (p) and marginal costs (c'), but this equation suggests an alternative based on θ , ε , and HHI . Since information on the market shares of individual firms is generally readily available, calculation of the Herfindahl-Hirschman index (HHI) is straightforward. This can be combined with an econometric estimate of the price elasticity of demand (ε), and an assumption about the degree of competition in a market (i.e. the conjectural variation θ), to calculate the Lerner index.

Econometric estimation of price elasticities is comparatively simple for homogeneous goods. In the case of differentiated products, it is more difficult due to the large number of substitute goods. If several competing firms each supply a number of substitutes, the number of cross-price elasticities that must be accounted for in the estimation can easily exceed the number of available observations. The market for breakfast cereals, for example, includes more than 50 different brands. Since the uncompensated cross-price elasticities between two brands do not have to be symmetric, it would be necessary to estimate 2,500 own- and cross-price elasticities for this market. This is referred to as the **curse of dimensionality**. The challenge for empirical researches is to formulate flexible demand systems that require the estimation of fewer parameters (such models are sometimes referred to as parsimonious).

Multi-stage models (hierarchical decision models) are one way to reduce the complexity of an empirical model. These models assume that households make sequential consumption decisions. At the top stage, for example, a household decides which share of its budget to allocate to breakfast cereals (relative to other foods). At the next stage, this budget share is sub-divided into different categories of cereals (for children, adults, organic products, etc.). The final stage then models the decisions to purchase individual brands within each category. In this way, the number of substitution parameters to be estimated is reduced significantly. At the same time, it is still possible to represent all substitution possibilities between brands within individual categories. Case study 9 describes an empirical application of this approach for the US beer market.

Case study 9: Market power on the US beer market (1)

Hausman et al. (1994) estimate price elasticities on the US beer market in a three-stage approach. In the first stage, the authors use a double-logarithmic demand function to estimate the price elasticity of demand for beer altogether. They estimate a value of -1.37 . In the second stage, the authors distinguish between three segments of the beer market ('premium', 'popular' and 'light'). For each segment, they calculate own- and cross-price elasticities again using a double-logarithmic demand function and assuming that total demand for beer is constant. The point estimates for the own-price elasticities are -2.7 for 'premium' and 'popular beer' and -2.4 for 'light beer'. In the third stage, they estimate the demand for various brands (and thus the substitution relationships within each of the three segments) using an Almost Ideal Demand System. The estimated own-price elasticities differ between the various brands of beer and range from -3.5 to -5.0 . The results of this study show that estimated own-price elasticities of demand depend on the level of aggregation that is studied. The higher the level of aggregation (e.g. beer as a whole compared with individual brands of beer), the fewer the consumers' substitution possibilities and, thus, the lower the estimated elasticities.

An alternative response to the curse of dimensionality in demand analysis is to assume that substitution relationships between products depend on the similarity of their characteristics. Rojas' (2008) study of the American beer market in case study 10 provides an example of this approach.

Case study 10: Market power on the US beer market (2)

Rojas (2008) uses a natural experiment to gain insights about the intensity of competition on the US beer market. In 1991, the sales tax on domestic and imported beer in the United States was doubled. The author analyses consumer demand for beer using 20 observations of quarterly price and quantity data for 64 different brands of beer in 58 US cities between 1988 and 1992. Rojas applies a so-called distance metric approach to reduce the number of substitution possibilities in this market. He uses information such as alcohol content, geographical distribution, product segment (light beer, lager, etc.) and packaging size to characterise each brand, and calculates the distances between brands in characteristics space. The distance between two brands i and j in n -dimensional characteristic space is $D_{ij} = [\sum_{n=1}^N (i_n - j_n)^2]^{\frac{1}{2}}$, where i_n (j_n) represents the n^{th} characteristic of product i (j). Rojas assumes that the cross-price elasticity of demand between products i and j

is a linear function of the distance between these products in characteristics space (D_{ij}). Using the resulting estimates of own- and cross-price elasticities, Rojas calculates Lerner indices in a conjectural variation framework. Regarding the practicability of this empirical approach, Rojas (2008 p. 26) concludes: "As in previous work, the inference conducted in this paper depends crucially on the precision of demand estimates. The distance metric method employed here is effective in reducing the number of ... cross-price effects, but it relies heavily in the researcher's ability to have data on all product characteristics that effectively determine substitution effects. Results may change if there are important unobserved product characteristics".

8.7.6. Market analyses in structural form

The analysis of structural, simultaneous models of supply (cost) and demand is the best option for identifying market power, provided that the necessary data are available. The so-called **New Empirical Industrial Organisation** (NEIO) literature frequently applies structural models to measure firms' market power, but recent empirical studies in business economics also rely on this approach to analyse the effects of different firm-level marketing activities.

Case study 11: NEIO on the banana market

Deodhar and Sheldon (1995) is one of the first applications of structural market analysis in the NEIO literature. They analyse the period between 1970 and 1992, when a few companies dominated the world market for bananas. United Brands, Standard Fruit and Noboa, for example, controlled more than 70% of the market in Germany. Deodhar and Sheldon use a simple NEIO model to measure these firms' market power on the German banana market.

The starting point is equation (11). In the case of n identical firms with constant marginal costs (c'), this equation becomes $p(Q) = -\frac{\partial p(Q)}{\partial Q}(1 + \theta)\frac{Q}{n} + c'$. The authors assume a linear market demand function: $Q = \alpha_0 + \alpha_1 p + Z\beta$, where Z is a vector of exogenous factors. The authors further argue that the supply of bananas on the German market is characterised by constant marginal costs: $c' = \gamma_0 + X\delta$, where X is a vector of exogenous supply-shifting factors. Inserting marginal costs into the price equation, we get: $p(Q) = -\frac{\partial p(Q)}{\partial Q}(1 + \theta)\frac{Q}{n} + \gamma_0 + X\delta = \gamma_1 Q + \gamma_0 + X\delta$, with $\gamma_1 =$

$$-\frac{\partial p(Q)}{\partial Q} \frac{(1+\theta)}{n} = -\frac{1}{\alpha_1} \frac{(1+\theta)}{n}$$
 (using $\alpha_1 = \frac{\partial Q}{\partial p}$ from the demand function). The authors estimate demand and supply functions based on annual observations for the period 1970 to 1992 to obtain point estimates for $\alpha_0, \alpha_1, \gamma_0, \gamma_1$, and the vectors β and δ . Specifically, the authors report the following estimates: $\alpha_1 = -0.32$ and $\gamma_1 = 0.91$. From this they calculate an indicator of the intensity of competition on the German banana market: $\lambda = \frac{1+\theta}{n} = -\alpha_1 \gamma_1 = 0.29$. Assuming that there are three (dominant) firms in the market ($n = 3$), this leads to an estimate of the conjectural variation θ of -0.13 . Hence, the firms' behaviour on this market differs significantly from perfect competition ($\theta = -1$) and from collusive behaviour (for $n = 3$, collusive behaviour would imply $\theta = n - 1 = 2$). However, it is not significantly different from Cournot-Nash behaviour ($\theta = 0$). Deodhar and Sheldon (1995 p. 345) conclude: "... the hypothesis of perfect competition and collusive behaviour could be rejected, but the hypothesis of Cournot-Nash behaviour could not be rejected".

Digal and Ahmadi-Esfahani (2002), Sheldon and Sperling (2003) and Perekhozhuk et al. (2017) provide surveys of numerous empirical applications of the NEIO approach for a wide range of agri-food markets. Kim and Cotterill (2008), which focuses on the cheese market, is an especially interesting application because it compares the results obtained from a structural model with those of a reduced form model, and thus highlights the additional insights into the exercise of market power that structural models are capable of generating.

In this section we have provided only a brief overview of the comprehensive literature on different approaches to measuring market power. Research on these issues is still very active, not least because robust empirical results on the existence and scope of market power are of great importance for competition policy. In the final section of this chapter, we will discuss some important aspects of competition policy and the institutions that implement it.

8.8. Competition policy and institutions

8.8.1. The importance of competition policy and the institutional framework

In a modern market economy, **competition policy** is one of the cornerstones of economic policy.¹² From medieval guild rules, to laws against cartels and monopolies, competition policy can be traced back to the Codex Hammurabi.¹³

Compared with other areas of economic policy, competition policy is very integrated in the EU. This is largely due to the fact that much economic activity in the EU takes place at a supranational level, i.e. many firms do business across borders within the EU. As a result, many questions and conflicts that arise in the area of competition policy also extend beyond national borders in the EU.

A special feature of competition policy in the EU is that not only the legislative body but also the executive authority is located at European level. The EU Commission proposes legislation and, together with the competition authorities of the member states, also enforces EU competition rules. The Commission's Directorate-General for Competition is the authority that is primarily responsible for these executive tasks. The Commission is also the investigative authority, prosecutor and decision-making authority in the first instance. The Directorate-General for Competition exercises broad powers in the assessment of subsidies by member states, in merger proceedings, and in antitrust investigations. These powers also affect multinational companies with head offices outside the EU that do business in the EU. The Commission's decisions can be challenged in the European courts.

Because they are so important for the design and functioning of the EU common market, the basic provisions of European competition law are stated in the Treaty on the Functioning of the European Union (Art. 101 et seq. TFEU). Article 101 TFEU prohibits horizontal and vertical "agreements between undertakings, decisions by associations of undertakings and concerted practices which may affect trade between

¹² Motta (2004) and Bishop and Walker (2009) provide detailed discussions of key topics in modern competition policy.

¹³ The Codex Hammurabi is the oldest completely preserved collection of legal acts and was enacted in Babylon in the 18th century BC.

Member States, and which have as their object or effect the prevention, restriction or distortion of competition within the internal market". This includes, for example, price-fixing agreements between firms, and arrangements whereby firms agree to sell their products or purchase inputs only within demarcated territories.

Article 102 TFEU stipulates that any "abuse by one or more undertakings of a dominant position within the internal market or in a substantial part of it shall be prohibited as incompatible with the internal market in so far as it may affect trade between Member States". A violation of Article 102 would be, for example, if a dominant firm were to sell its products below costs with the objective of driving competitors out of the market. Merger controls also fall under Article 102 TFEU.

In the EU, competition policy is implemented both by the EU and by national competition authorities. The EU is responsible for cases with supra-national implications while national authorities are responsible for cases that do not affect trade between the member states. Since it is not always possible to sharply divide jurisdictions, EU and national authorities sometimes work on a case simultaneously. However, merger controls are always pursued exclusively at one level or the other, as we discuss in the next section.

8.8.2. Subfields of competition policy

Competition policy in the EU (and in most developed economies) rests on three pillars: **abuse supervision**, **antitrust supervision**, and **merger control**. The first two pillars (abuse and antitrust supervision) are predominantly *ex post*-oriented. In both of these areas, the competition authority only becomes active when economic damage has occurred (or is suspected). Merger control is *ex ante*-oriented and seeks to prevent firms from acquiring dominant positions on markets.

a) Ex post approaches: abuse and antitrust supervision

A proceeding against a horizontal or vertical restriction of competition is initiated by an application from an eligible party. Such an application must include a specific demand, for example that firms stop fixing prices. In practice, the competition authorities themselves conduct investigations and collect evidence on the possible abuse of market power or antitrust behaviour by firms. Firms are legally required to cooperate with such investigations, and they can be fined for failure to do so.

Abuse supervision is often considered a comparatively weak instrument of competition policy. Illegal cartels and price-fixing take place in secret, often on the basis of verbal agreements and without documentation. Hence, they are difficult to detect and prove. Insider knowledge is often the key to detecting illegal cartels. Hence, competition laws often contain provisions that allow the competition authority to waive or reduce penalties for whistle-blowers who provide evidence that leads to the detection of a cartel in which they have participated. In Germany, for example, about half of all antitrust proceedings investigated by the Bundeskartellamt (Federal Cartel Office) are triggered by whistle-blowers. In addition, whistle-blower provisions create uncertainty among the members of a cartel; they undermine trust and thus the stability of a cartel. In an empirical analysis of the effects of the whistle-blower provisions introduced in EU competition law in 1996, Brenner (2009) evaluates 61 antitrust decisions reached by the European Commission between 1990 and 2003, and finds that the whistle-blower provisions led to significantly increased penalties and shorter proceedings. Miller (2009) provides a similar empirical analysis for the US.

b) Ex ante approach: Merger control

Of the three pillars of competition policy, merger control is perhaps the most important; it aims to prevent firms from acquiring market power through mergers. Planned mergers must be notified to the competition authority if the annual turnover of the participating firms exceeds a given threshold, and a planned merger cannot be completed as long as the authority is carrying out its investigations. The turnover thresholds also determine whether a merger control is carried out by the EU competition authority or at the national level. The competition authority for its part must reach a decision to permit or forbid a planned merger within a maximum length of time (which is uncommon in other areas of legal practices).

In national merger control cases in Germany, the competition authority has one month to decide in Phase I (pre-investigation) whether a planned merger should be permitted or is subject to further investigation. If there is reason to believe that the planned merger could have negative implications for competition, the competition authorities open Phase II (main investigation). Phase II is an in-depth analysis of the merger's effects on competition. The competition authority must within 4-5 months either forbid the merger or approve it, sometimes subject to conditions. Completing a merger without permission can lead to financial penalties for the participating firms, and invalidation of the merger.

Because the time available is limited, especially in Phase I, the competition authority is not in a position to carry out detailed empirical analyses. For a first rough assessment of the price effects of a merger, the conjectural variation model (Section 8.4) is useful. According to equation (14), the aggregate Lerner index for a market in equilibrium is $L = \frac{1+\theta}{\varepsilon} HHI$. If a merger does not affect firm conduct (represented by parameter θ), and if the price elasticity of demand (ε) remains unchanged, then a merger will affect the price mark-up (the Lerner index L) by changing the market shares of the firms in the industry and, thus, the Herfindahl-Hirschman index (HHI). The difference between the pre- and post-merger HHI in an industry is sometimes referred to as the ‘delta’.

In its *Guidelines on the assessment of horizontal mergers*, the European Commission (2004, p. 7) provides some guidance for assessing horizontal mergers with the HHI . "The Commission is unlikely to identify horizontal competition concerns in a market with a post-merger HHI below 1 000. Such markets normally do not require extensive analysis. The Commission is also unlikely to identify horizontal competition concerns in a merger with a post-merger HHI between 1 000 and 2 000 and a delta below 250, or a merger with a post-merger HHI above 2 000 and a delta below 150, except where special circumstances ... are present", which are further explained in the Guidelines.

However, a fundamental condition for the calculation of the HHI (and for making informed decisions on competition issues in general) is accurate information on the market shares of the firms in an industry. This requires an accurate definition of the relevant market, which plays a central role in many cases that are considered by competition authorities.

8.8.3. Criteria for defining the relevant market

Determining whether a firm is abusing market power, or whether a merger will give a firm a dominant position on a market, critically depends on how narrowly (or broadly) the relevant market is defined. Firms that want to merge understandably have an interest in defining their market as broadly

as possible¹⁴, while others who want to prevent the merger will benefit from a narrow market definition.

Competition policy defines the relevant market in two dimensions; geographic space and product space. “The objective of defining a market in both its product and geographic dimension is to identify those actual competitors of the undertakings involved that are capable of constraining those undertakings’ behaviour and of preventing them from behaving independently of effective competitive pressure” (European Commission 1997, paragraph 2). The relevant behaviour in this context is price setting, and the crucial question is the extent to which a firm’s ability to raise prices is constrained by competing firms. Hence, the definition of the relevant market and market power are closely intertwined.

As an example of geographic market definition consider a situation in which there are two firms, *A* and *B*, which both supply an identical product but are located at different places. Assume further that there are costs associated with transporting the product to customers. If the geographic distance between *A* and *B* is so large that *B*’s activities (such as choosing the product price) do not affect the competitive situation for *A*, then we can conclude that the markets are (geographically) separate, and *A* can behave as a monopolist on its market. However, if *B*’s activities influence the competitive situation for *A* (for example, because transport costs fall so that *A*’s and *B*’s markets begin to overlap), *A* can no longer behave as a monopolist. *B*’s activities now have a disciplining effect on firm *A*, which can no longer raise its prices without fear of losing customers to *B*. In this case, the two firms operate in the same (geographic) market.

A similar definition underlies the definition of the relevant market in product space: “A relevant product market comprises all those products and/or services which are regarded as interchangeable or substitutable by the consumer, by reason of the products’ characteristics, their prices and their intended use” (European Commission 1997, paragraph 7).

The relevant market is based on a combination of the geographic and the product dimensions, and the following three criteria:

¹⁴ An example of a very broad definition of the relevant market is the definition suggested by the Coca-Cola company: “*The company’s (Coke) executives are fond of saying that every human needs to consume an average of 64 ounces of fluid each day ... to survive. Of that total, Coke calculates it now accounts for less than two ounces.*” (Wall Street Journal, May 8, 1997).

a) Product characteristics

The Commission is very cautious about this criterion for market definition. "An analysis of the product characteristics and its intended use allows the Commission, as a first step, to limit the field of investigation of possible substitutes. However, product characteristics and intended use are insufficient to show whether two products are demand substitutes. Functional interchangeability or similarity in characteristics may not, in themselves, provide sufficient criteria, because the responsiveness of customers to relative price changes may be determined by other considerations as well." (European Commission 1997, paragraph 36). Regarding the "responsiveness of customers to relative price changes", a distinction is made between substitutability on the supply and on the demand side.

b) Demand-side substitutability

The EU competition authority correctly attaches the greatest importance to demand-side substitutability. Substitution by customers is the most immediate and effective disciplining factor for a firm. Competition authorities analyse which alternatives (substitutes) exist for a product, and to what extent customers would switch to these substitutes if the product's price were to rise. It is not necessary that all customers switch to an alternative product for it to be considered a substitute; what matters is whether a sufficient number of consumers switch. Those consumers who are most likely to switch are referred to as marginal consumers. If a sufficient number of consumers is likely to switch to a substitute, then it will not make sense for a firm to increase its price.

c) Supply-side substitutability

A firm's price-setting behaviour will also be constrained if there is sufficient supply-side substitutability. Supply-side substitutability is given when competitors are able to react to a price increase for a product by increasing the supply of that product quickly and without incurring significant additional costs or risks. The terms "quickly" and "significant additional costs or risks" are admittedly vague and require case-by-case elaboration. Supply-side substitutability plays an important role in practice, especially

when suppliers offer a wide range of differentiated products.¹⁵ The quantitative importance of supply-side substitutability is reflected in statistics on product innovation. For example, a large share of the product innovations in the German food industry can be considered so-called me-too innovations, i.e., the attempt by firms to enter their rivals' profitable market segments.

8.8.4. Instruments and methods for defining the relevant market

Numerous instruments and methods for defining the relevant market have been proposed in the economic literature. The European Commission states that it “does not follow a rigid hierarchy of different sources of information or types of evidence” but rather “follows an open approach to empirical evidence, aimed at making an effective use of all available information which may be relevant in individual cases” (European Commission 1997, paragraph 25). In many cases important data are missing and the competition authorities have to work with approximations and indirect estimates.

The methods for defining relevant markets that we will discuss below are often also used by competition authorities when they investigate possible abuse of market power. In the following we mainly focus on application to defining the relevant market; applications to the measurement of market power and the abuse of market power are addressed separately.

¹⁵ For example, shoes of different sizes clearly belong to the same market. From an individual customer's point of view, switching from size 8 to size 6 would entail significant costs (physical pain); demand-side substitution is therefore limited. However, shoe producers can adjust the production of shoes of different sizes in a short time period without incurring significant additional costs or risks. Hence, supply-side substitutability is given.

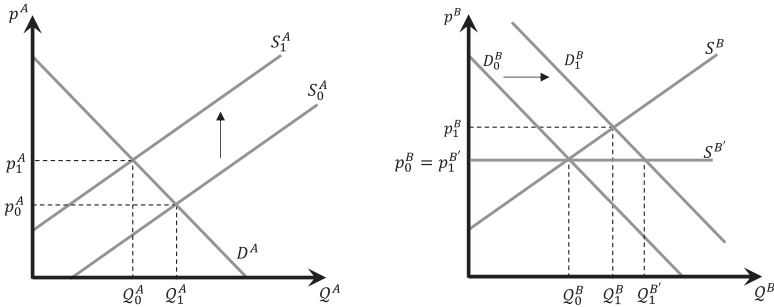
a) Price-based procedures for defining the relevant market

Many economists refer to the **law of one price** (LOP) when determining the extent of a market.¹⁶ The test on uniform price development proposed by Horowitz (1981) builds on this idea. The LOP is relaxed somewhat to account for transfer costs between locations and possible random disturbances, but even after allowing for these effects, prices at different locations will tend to move in parallel with a fixed margin between them if the locations are part of the same market. To quote Horowitz (1981, p. 8): "if two geographic areas ... encompass a single market for a particular product, then over time the price at which the product exchanges in one area cannot remain continually out of line with the price at which it exchanges in the other".

In empirical applications, partial correlations between prices of different products are often calculated in order to determine whether these products belong to the same market. If two products A and B are close substitutes, then their prices should exhibit a strong and positive correlation. For example, if, in Figure 8.8, an increase in the marginal cost of product A leads to a shift in the supply curve from S_0^A to S_1^A , the price increase for product A will lead to increased demand for product B ; the demand curve for product B will shift from D_0^B to D_1^B . Given the supply curve S_B (for product B), a price increase will also be observed for product B (from p_0^B to p_1^B). The positive correlation between the prices of A and B will increase with the degree of substitutability between the two products.

¹⁶ The LOP states that under perfect competition and in the absence of any barriers to trade, identical goods in different locations will be sold for identical prices. If the prices for identical goods in different locations are not identical, then the two locations are not part of the same market. „It is well known that by market economists mean, not a certain place where purchases and sales are carried on, but an entire territory of which the parts are so united by the relations of unrestricted commerce that prices there take the same level throughout, with ease and rapidity” (Cournot 1897, S. 51-52).

Figure 8.8: Price-based definition of the relevant market



Source: Own depiction.

In more recent applications, advanced time series methods (Granger causality tests and cointegration methods) rather than simple correlation coefficients are used to measure the extent of a market. These methods are better able to determine the direction of causality between prices, to capture the dynamic relationships between prices, and to distinguish between short- and long-term price adjustments.¹⁷ Such price-based approaches have several advantages. They are comparatively simple to implement and to interpret (which can be especially important in court proceedings). In addition, they require only price data, which are often readily available.

However, price-based approaches also have important disadvantages. In many markets, list prices differ from actual transaction prices. Furthermore, correlation analysis can produce different results for the same market depending on the frequency of the price data that are used. In addition, distinguishing between true and spurious correlation is sometimes difficult: prices for two products can be (spuriously) correlated even if there are no supply-side or demand-side substitution relations between them, for example if both are subject to the same inflation or business cycle fluctuations. Moreover, there are no objective benchmark values that can be used to determine what constitutes a ‘large’ or ‘small’ correlation between two prices.

¹⁷ These time series methods are also used to analyse relationships between prices at different levels of the food chain. These price relationships are often referred to as **vertical**, while price relationships between locations are referred to as **spatial** or **horizontal**. We discuss vertical price relationships and also provide a brief introduction to the relevant time series methods in Chapter 10.

Finally, the results of price-based analyses are sometimes difficult to interpret, as illustrated in Figure 8.8. An increase in the price of product A (from p_0^A to p_1^A), and the resulting increase in demand for product B (from D_0^B to D_1^B , due to demand-side substitutability), will only trigger an increase in the price of product B if the supply of B is not perfectly elastic. This will be the case if the supply-side substitutability between A and B is relatively weak. However, if suppliers can shift their production from A to B quickly and without incurring significant additional costs or risks (i.e. if there is strong supply-side substitutability between A and B), then the supply of B will be elastic and the correlation between the prices for A and B will be low. In the extreme case in which the supply of B is perfectly elastic (supply curve $S^{B'}$ in Figure 8.8), p^B will remain constant ($p_0^B = p_1^{B'}$). Hence, an empirical analysis would find a weak (or no) correlation between the prices of the two products and thus conclude that A and B are weak substitutes, although they are actually characterised by high supply-side substitutability.

The main problem with purely price-based attempts to define the relevant market is that price levels and price relationships between markets depend on supply and demand conditions (the shapes of the supply and demand curves) on those markets. However, information on consumer behaviour (reflected in the price elasticity of demand) and on production technologies and costs (reflected in the price elasticity of supply) are not included in purely price-based analyses.

b) The SSNIP Test (HM Test)

The SSNIP (small but significant non-transitory increase in price) or HM (hypothetical monopoly) test was incorporated into the merger guidelines of the US Department of Justice in 1984. It also plays a central role in the European Commission (1997) Guidelines for defining and delineating relevant markets. This test has become one of the most important instruments for defining the extent of markets in merger cases.

The following example illustrates the procedure. In order to determine whether two products A and B belong to the same market, we assume that product A is supplied by a (hypothetical) monopolist. The SSNIP test examines whether it would be profitable for this monopolist to increase the price of A from its current level by 5 to 10%. If such a price increase is profitable because the quantity of A demanded does not fall substantially, then A and B can be considered separate markets. If, however, the price increase is not profitable, because many consumers do switch to product B

(or simply stop purchasing product *A* altogether), then *A* and *B* are considered to belong to the same market.

If more than two products must be considered, the test proceeds iteratively. The test starts from the smallest relevant market (candidate market) and adds further products (the next best substitutes) to the candidate market step-by-step. The relevant market is defined as the narrowest set of products for which it would be profitable for a hypothetical monopolist to impose a small but significant and permanent price increase.

Whether a price increase is actually ‘small but significant’ is subject to interpretation and depends on a number of factors such as the magnitude of typical price fluctuations on the markets in question, and the rate of inflation. The US Department of Justice uses a 5% price increase, while the European Commission investigates the effects of 5 to 10% price increases, depending on the product. It is also not entirely clear what period of time should be used when testing for an increase in profits with the SSNIP test. In the very short run, consumers will not be able to search for possible substitute products, and other enterprises will not be able to enter the market with their competing products. In the very long run, consumers and competitors will be able to react and it will be difficult for any hypothetical monopolist to generate excess profits via inflated prices. In practice, competition authorities work with a period of one year, however there is no compelling economic rationale for this choice.

A key element of the SSNIP test is determining when it would not be profitable for a firm to increase prices. Following a price increase, some customers will stop consuming the firm’s product altogether, while others will switch to rival products (demand-side substitution). At the same time, competing firms will increase their production or launch entirely new substitute products (supply-side substitution). If demand- and supply-side substitution effects are strong, demand will be price-elastic and the price increase will lead to a large reduction in the firm’s sales. As the firm sells less, it will reduce production and its costs will fall as well.

The competition authorities therefore focus on measuring the demand and cost changes that result from a price increase. How high can the price elasticity of demand be so that a price increase is just barely (un)profitable? The answer to this question is sometimes referred to as the critical demand elasticity (Werden 1998), as illustrated in case study 12.

Case study 12: The SNIPP test on the beer market

In March 2016, competition authorities of various countries were notified about the proposed acquisition of the world's second largest brewery (SABMiller) by the world's largest brewery (Anheuser-Busch InBev). Competition authorities examined and finally approved this acquisition (see, among others, the European Commission, 2016). The first step involved a detailed analysis of the relevant market (in terms of geographical and product space). Gürkaynak and Kalkan's (2017) conducted a SSNIP test to determine the relevant market in Turkey. Based on monthly sales data for 145 products over the period from March 2013 to March 2016, they estimated own- and cross-price elasticities of demand. For the premium segment of the market, they found an own-price elasticity of retail demand of -7.99 (this figure is much higher than Hausmann et al.'s (1994) estimate for the US beer market - see case study 9). They found that the average own-price elasticity of demand for an individual brand is -12.83 . At the wholesale level, the authors find a lower price elasticity of -3.47 . This result suggests that the demand for beer is price elastic even at the wholesale level, and that the scope for brewers to raise prices is therefore limited. The authors then compare the actual elasticity with a critical demand elasticity (ε^{krit}), defined as the elasticity value at which the consumer response would be just elastic enough to render a price increase of $\alpha\%$ unprofitable for the seller (in this case brewer). The authors use a formula proposed by Werden (1998) to calculate the critical elasticity: $\varepsilon^{krit} = \frac{100}{1+\alpha}$. In their application, the Lerner index for 2015 is $L = 46.6$. For $\alpha = 5$ or $\alpha = 10$, they find that ε^{krit} is smaller than the estimated elasticity at the wholesale level (in absolute values): $\varepsilon_{\alpha=5}^{krit} = \frac{100}{(46.6+5)} = 1.94 < \varepsilon = 3.47$ or $\varepsilon_{\alpha=10}^{krit} = \frac{100}{(46.6+10)} = 1.77 < \varepsilon = 3.47$ "It has been shown that actual elasticity of demand for the premium segment is higher than the critical elasticity ... These findings lead to the conclusion that the relevant product market regarding beer brands in Turkey must be wider than the premium segment" (p. 345).

The application of the SSNIP test is comparatively complex and data-intensive. Beginning with the smallest relevant market, products are added one-by-one and the SSNIP test repeated for each iteration. At each step, own and cross-price elasticities of demand and the Lerner index must be estimated in order to determine whether a price increase would be profitable. Carrying out this procedure often requires more time than is available. In addition, the marginal costs of the relevant firms are generally

unknown, and price elasticities of demand are often difficult to estimate reliably.

In view of these difficulties and since competition authorities must often reach decisions within short periods of time, practical applications of the SSNIP test often depart from the procedure described above. For example, the selection of the products that are added to the candidate market is often made on an *ad hoc* basis and the SSNIP test is only carried out for those products that are considered particularly interesting based on *a-priori* considerations. Rather than estimating demand elasticities econometrically, they are roughly assessed based on readily available information. In addition, the application of the test can be further simplified by making assumptions about the conduct of firms on the market under consideration. As shown in Section 8.4, in a Cournot model ($\theta = 0$) with identical suppliers the Lerner index simplifies to $L = \frac{p-c}{p} = \frac{1}{\varepsilon}r$, where r represents firms' market shares. If one believes that this model is a reasonable approximation of reality, then the Lerner index can be calculated without information on marginal costs by using the price elasticity of demand and the firms' market shares.

The SSNIP test is well established as tool for defining the extent of the relevant market. However, it has an important weakness as an instrument for identifying the abuse of market power. The problem is known as the **cellophane fallacy**, a term that refers to a 1956 US Supreme Court decision on whether the company *du Pont* held a dominant market position and was abusing the associated market power. According to a narrow definition of the relevant market that considers only the product cellophane, *du Pont* had a market share of about 70%. According to a broader definition argued by *du Pont*, other packaging materials (such as aluminium foil, wax paper and polyethylene) are close substitutes for cellophane, and *du Pont*'s market share was only roughly 17%, in which case it did not have a dominant market position. The Supreme Court based its decision to define the market broadly on estimates of cross-price elasticities; high cross-price elasticities indicated that consumers consider alternative packaging materials as sufficiently close substitutes for cellophane.

This case highlights an important weakness of the SSNIP test as an instrument for identifying market power abuse. In the case of a merger, the possible creation of market power is examined *ex ante*. However, the possible abuse of market power is assessed *ex post*. If a firm is abusing

market power, then the price it charges will lie above its marginal cost, and it will already have implemented any profitable price increases.

As a result, the estimation and interpretation of the price elasticity of demand in the application of the SNNIP test can be misleading. Unless the demand function is iso-elastic, the price elasticity of demand will vary with price p . This is particularly evident in the case of a linear demand function, where ε is directly proportional to p/q . While the market price and the price elasticity of demand will be low in a competitive market, the price elasticity of demand will typically be high in a monopoly. In this case, therefore, a high price elasticity of demand will be the result of the monopolist's abuse of market power and cannot be interpreted as indicating the absence of market power, as a mechanical application of the SSNIP test would suggest.

Hence, in a highly concentrated market, the prevailing price elasticity of demand cannot be used to define the extent of the relevant market. The European Commission acknowledges this issue as follows: "Generally, and in particular for the analysis of merger cases, the price to take into account will be the prevailing market price. This may not be the case where the prevailing price has been determined in the absence of sufficient competition. In particular for the investigation of abuses of dominant positions, the fact that the prevailing price might already have been substantially increased will be taken into account." (European Commission 1997, paragraph 19). However, the extent to which a market is classified as concentrated itself depends on how the extent of the market is defined in geographic and product space.

This brief overview of competition policy illustrates the close links between economic theory and policy: the empirical analysis that is used in applied competition policy is based on solid microeconomic models. However, we have also seen that for some problems (such as defining the extent of the relevant market) methods that are both theoretically well founded and meet the practical requirements of applied competition policy are still lacking. As ongoing globalisation leads to increasing horizontal and vertical integration of production processes, defining the extend of the relevant market will continue to pose a major challenge.

8.9. Keywords

- Abuse and antitrust supervision
- Cellophane fallacy
- Competition policy
- Conjectural variation
- Cournot-Nash equilibrium
- Curse of dimensionality
- Indistinguishability theorem
- Law of one price
- Lerner index
- Mark-up
- Merger control
- Mixed oligopoly
- Nash-Bertrand equilibrium
- New Empirical Industrial Organisation
- Perceptual maps
- Predatory and cooperative advertising
- Strategic complements and strategic substitutes
- Repeated games
- Spatial and vertical price relationships
- Structure-conduct-performance
- Super games

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Appendix 8.1 Derivation of (inverse) demand functions for n goods

To maximize the quadratic utility function

$$U(q, q_0) = \sum_{i=1}^n s_i q_i - \frac{1}{2} \left(\sum_{i=1}^n q_i^2 - 2d \sum_{j=1}^n q_i q_j \right) + q_0$$

for $i \neq j$ with respect to the budget constraint

$$y = \sum_{i=1}^n p_i q_i + q_0$$

we formulate the Lagrangian equation:

$$L = \sum_{i=1}^n s_i q_i - \frac{1}{2} (\sum_{i=1}^n q_i^2 - 2d \sum_{j=1}^n q_i q_j) + q_0 + \lambda (y - \sum_{i=1}^n p_i q_i - q_0)$$

The first derivatives with respect to q_i , q_0 and λ are:

$$\frac{\partial L}{\partial q_i} = s_i - q_i + d \sum_{j=1}^n q_j - \lambda p_i = 0 \quad (15)$$

$$\frac{\partial L}{\partial q_0} = 1 - \lambda = 0 \quad (16)$$

$$\frac{\partial L}{\partial \lambda} = y - \sum_{i=1}^n p_i q_i - q_0 = 0 \quad (17)$$

Note that the Lagrange multiplier (λ), which measures the shadow price of income, is constant ($\lambda = 1$ in equation 16). This allows us to conduct a partial equilibrium analysis of the n goods, since there are no interaction (income) effects with other goods (represented by the composite commodity q_0). For $\lambda = 1$, the inverse demand equation for the n goods from equation (15) is: $p_i = s_i - q_i + d \sum_{j=1}^n q_j$. This corresponds to equation (2) above. The demand for the composite commodity q_0 is given by equation (17): $q_0 = y - \sum_{i=1}^n p_i q_i$.

To derive equation (3) from the inverse demand equation (2), we aggregate equation (2) over all n goods:

$$\sum_{i=1}^n p_i = \sum_{i=1}^n s_i - \sum_{i=1}^n q_i - d \sum_{i=1}^n \sum_{j=1}^n q_j,$$

which is equivalent to

$$\sum_{i=1}^n p_i = \sum_{i=1}^n s_i - \sum_{i=1}^n q_i - d(n-1) \sum_{i=1}^n q_i$$

because $\sum_{i=1}^n \sum_{j=1}^n q_j = (n-1) \sum_{i=1}^n q_i$, for $i \neq j$.¹⁸

This gives $\sum_{i=1}^n q_i = \frac{\sum_{i=1}^n s_i - \sum_{i=1}^n p_i}{1-d(n-1)}$. Inserting this into equation (2) we obtain

$$q_i = s_i - p_i + d(\sum_{i=1}^n q_i - q_i)$$

which can be transformed into

$$q_i(1+d) = s_i - p_i + d \frac{\sum_{i=1}^n s_i - \sum_{i=1}^n p_i}{1-d(n-1)}$$

or

$$q_i(1+d) = s_i - p_i + d \frac{\sum_{j=1}^n s_j - \sum_{j=1}^n p_j}{1-d(n-1)} + d \frac{s_i - p_i}{1-d(n-1)}.$$

After further transpositions we obtain:

$$q_i(1+d) = \frac{[1-d(n-1)]s_i + d\sum_{j=1}^n s_j}{1-d(n-1)} - \frac{[1-d(n-1)]p_i + d\sum_{j=1}^n p_j}{1-d(n-1)} + \frac{d\sum_{j=1}^n p_j}{1-d(n-1)}.$$

Dividing by $(1+d)$ and rearranging terms leads to equation (3):

$$\begin{aligned} q_i &= \frac{[1-d(n-2)]s_i + d\sum_{j=1}^n s_j}{(1+d)[1-d(n-1)]} - \frac{[1-d(n-2)]p_i}{(1+d)[1-d(n-1)]} - \frac{d\sum_{j=1}^n p_j}{(1+d)[1-d(n-1)]} \\ &= \sigma_i - \beta p_i + \delta \sum_{j=1}^n p_j \end{aligned}$$

Appendix 8.2 Pricing in a two-product monopoly

In the following, we examine the profit maximisation problem of a monopolist who sells two differentiated products to customers. Consumer demand is given by equation (8.3). The monopolist maximises the total profit for all n markets: $\Pi^M = \sum_{i=1}^n \pi_i^M = \sum_{i=1}^n [(\sigma_i - \beta p_i + \sum_{j=1}^n \delta p_j)(p_i - c'_i) - f_i]$ for $i \neq j$. Here, the monopolist's marginal and fixed costs are represented by c'_i and f_i . For $n = 2$, the first-order

¹⁸ This can be illustrated for the case of $n = 3$. For $i \neq j$, $\sum_{i=1}^3 \sum_{j=1}^3 q_j = (q_2 + q_3) + (q_1 + q_3) + (q_1 + q_2) = (n-1) \sum_{i=1}^3 q_i$.

conditions are: $\frac{\partial \Pi^M}{\partial p_i} = \sigma_i - 2\beta p_i + \delta p_j + \beta c'_i + \delta(p_j - c'_j) = 0$ for $i, j = 1, 2$. From this, we obtain the following profit-maximising prices and quantities: $p_i^M = \frac{(\beta^2 - \delta^2)c'_i + \beta\sigma_i + \delta\sigma_j}{2(\beta^2 - \delta^2)}$ and $q_i^M = \frac{\sigma_i - \beta c'_i + \delta c'_j}{2}$. Profits are $\Pi^M = \pi_i^M + \pi_j^M = \frac{(\sigma_i - \beta c'_i + \delta c'_j)[\beta\sigma_i - \delta\sigma_j - (\beta^2 - \delta^2)c'_i]}{4(\beta^2 - \delta^2)} - f_i + \frac{(\sigma_j - \beta c'_j + \delta c'_i)[\beta\sigma_j - \delta\sigma_i - (\beta^2 - \delta^2)c'_j]}{4(\beta^2 - \delta^2)} - f_j$. For $c'_i = c'_j = c'$, $\sigma_i = \sigma_j = \sigma$ and $f_i = f_j = f$, this equation simplifies to $\Pi^M = \pi_i^M + \pi_j^M = 2 \frac{[\sigma - (\beta - \delta)c']^2}{4(\beta - \delta)} - 2f$. In the case of independent products ($\delta \rightarrow 0 \Rightarrow \beta = 1$), we get $p_i^M = \frac{\sigma + c'}{2}$, $q_i^M = \frac{\sigma - c'}{2}$ and $\pi_i^M = \frac{(\sigma - c')^2}{4} - f$ as well as $\Pi^M = 2\pi_i^M = \frac{(\sigma - c')^2}{2\beta} - 2f$. This corresponds to the standard one-product monopoly market with linear demand and cost functions, which can be found in every introductory textbook of microeconomics.

Exercises

1. The big agricultural machinery manufacturers spend considerable sums on advertisement and promotion, for example in agricultural magazines and trade fairs. Do they engage primarily in cooperative or predatory advertising?
2. Your consulting company estimates that the own-price elasticity of demand for a client's product is -0.6. The client is the only firm that is entitled to sell this product in a particular region. In order to raise profits from selling this product, you recommend that the client:
 - a) increase production,
 - b) lower the price,
 - c) reduce production,
 - d) choose output such that price equals marginal cost,
 - e) choose output such that price equals average costs.

Hint: Only one answer is correct.

3. In a detailed econometric analysis, Gasmi et al. (1992) estimate demand and cost functions for two competing firms (*Coca-Cola Comp.* and *Pepsico Inc.*) in the US soft-drink market for the period 1968 to 1986. On the basis of Table 11 (page 294) of their study, we obtain (after rounding the parameter estimates) the following demand functions:

$q_{CO} = 60 - 4p_{CO} + 2p_{PE}$ and $q_{PE} = 50 - 5p_{PE} + p_{CO}$, where the subscript *CO* denotes *Coca-Cola Comp.* and *PE* denotes *Pepsico Inc.*. Following Gasmi et al. (1992, Table 11) we also assume (constant) marginal production costs of $c_{CO} = 8$ and $c_{PE} = 7$.

- a) If *Pepsico Inc.* were to sell its product at a price of $p_{PE} = 8$, what profit-maximizing price p_{CO} should *Coca-Cola Comp.* (p_{CO}) choose under price (Bertrand) competition?
 - b) Compute the prices (p_{CO} and p_{PE}) and quantities sold (q_{CO} and q_{PE}) for both companies in Bertrand equilibrium.
 - c) Compute the Lerner-Index for both companies (L_{CO} and L_{PE}) as well as the Herfindahl-Hirschmann-Index (*HHI*) for this market in Bertrand equilibrium.
4. In an oft-cited study, Genesove and Mullin (1998) analyse the US sugar industry at the end of the 19th century. The technology for producing refined sugar was simple back then, which enables the authors to calculate the producers' marginal costs. Specifically, they find $c^L = 3.78$ and $c^H = 3.87$ (in cents per 100 pounds of sugar), where H denotes the high and L the low season. Based on the study (see Genesove and Mullin, 1998, Table 3 on p. 366) assume that the demand for sugar (which is a comparatively homogeneous produce) is given by the following linear functions $Q^L = 13.37 - 2.30p^L$ and $Q^H = 10.74 - 1.36p^H$, where prices (p^L and p^H) are in cents per 100 pound, and quantities (Q^L and Q^H) are in 100 pound.
- a) In 1892 the *American Sugar Refining Company* (ASRC) had a market share of 95%. On the assumption that this company was effectively the monopoly supplier of sugar in the US, what profit-maximizing prices should it have charged in the high and in the low season? What would have been the corresponding mark-ups (μ) and Lerner indices (L)?
 - b) The actual average price of sugar was $p^N = 3.99$ in the high and $p^H = 4.14$ in the low season. Calculate the actual mark-ups (μ) and Lerner indices (L) corresponding to these prices. What might explain the fact that the actual prices were considerably lower than the profit-maximising monopoly prices, even though the market was highly concentrated?
 - c) In the ensuing years up to 1914, other producers entered the US sugar market and ASRC's market share steadily fell. Assume that two firms entered the market, so that $n = 3$. In addition, assume for simplicity that all firms were identical. Using the information provided above on the demand function and the marginal costs in the

- high season, calculate the equilibrium prices for a conjectural variation model with (i) $\theta = 0$ and (ii) $\theta = -0.5$.
- d) Assume that the number of (identical) competitors was $n = 3$ and the actual price of sugar in the high season was $p^H = 4.14$. How high must the parameter of conjectural variation θ have been to generate this equilibrium outcome?
5. The so-called ABCD companies (*ADM*, *Bunge*, *Cargill* and *Louis Dreyfus*) control a large share of the international grain trade. What factors suggest that these firms will (will not) behave collusively? Consider Feuerstein's (2005) ten stylized facts when formulating your answer to this question. How might the growing importance of the Chinese state trading enterprise *Cofco* affect the conduct of the ABCD companies?